Scenario 1: Business Operations — AI Operations Copilot (Decision Support Only)

Phase 1: Problem Definition & Objectives

1. Primary User Persona

Persona: Business Operations Analyst / Product Manager

Profile:

- Works with sales, revenue, or operational datasets
- Uses dashboards (Tableau, Power BI, Excel)
- Needs quick insights for decision-making
- Not deeply technical (limited SQL/Python)

Pain Points:

- Too much data, not enough insights
- Manual analysis is slow
- Root cause analysis takes time
- Needs faster answers for stakeholders

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2. Daily Workflow (Current vs Proposed)

Current Workflow (Manual)

1. Open dashboards / Excel
2. Filter data manually
3. Compare time periods
4. Identify trends
5. Try to guess root cause
6. Prepare explanation

Problems

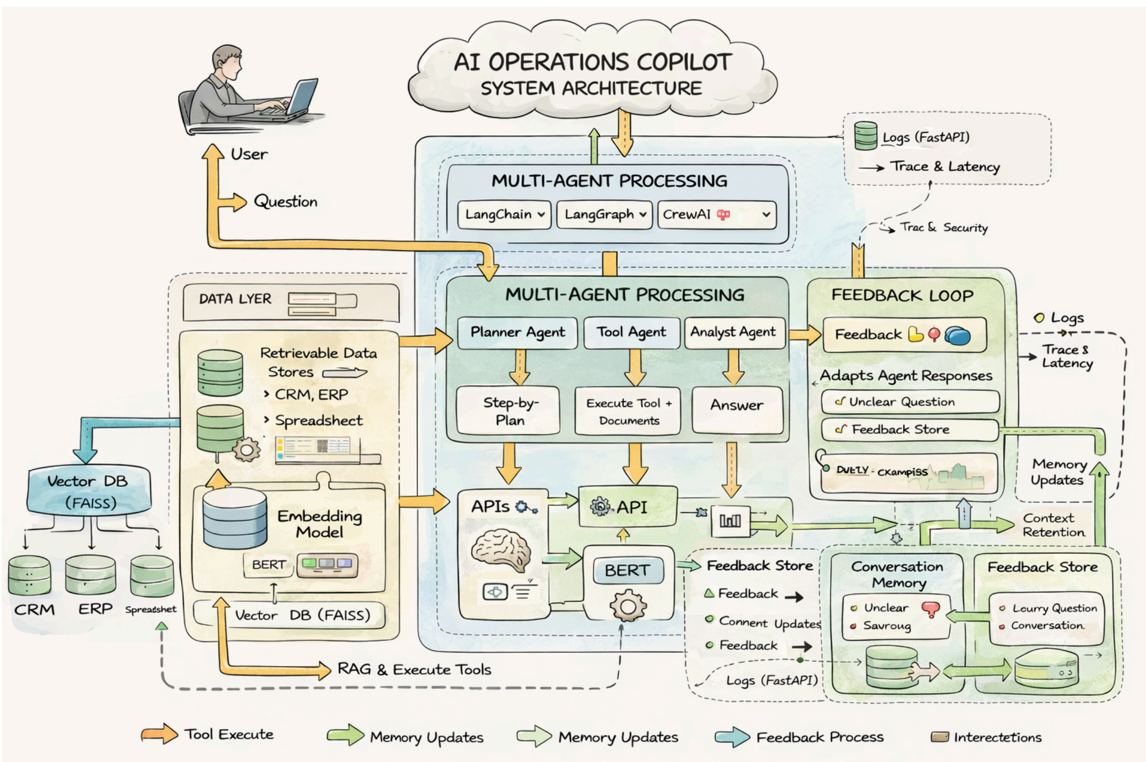
- Time-consuming
- Error-prone
- Requires expertise

Proposed Workflow (AI Copilot)

1. Upload dataset (CSV)

2. Ask question in natural language
   👉 *"Why did revenue drop last week?"*
3. System:
   - Retrieves relevant data
   - Analyzes trends
   - Generates explanation + confidence
4. User gets:
   - Final answer
   - Explanation
   - Trace (reasoning)
   - Latency

Proposed System Architecture



Outcome

- Faster insights
- Reduced manual effort
- Explainable AI decisions

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3. Problem Statement

Organizations struggle to quickly extract actionable insights from operational data due to reliance on manual analysis tools and lack of intelligent systems that can interpret data contextually and explain results.

Core Problem:

- No **intelligent assistant** for business data analysis
- Insights require **manual effort + expertise**
- Lack of **explainability and traceability**

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4. Inputs, Outputs, Constraints, Assumptions

Inputs

- User query (natural language)
- CSV dataset (uploaded via UI)
- Selected:
  - LLM model (e.g., GPT-4.1-mini)
  - Agent framework (LangGraph / LangChain / CrewAI)

Outputs

- Final Answer
- Explanation
- Confidence Level
- Reasoning Trace
- Latency (response time)

Constraints

Technical

- Depends on LLM accuracy
- Context limited by token size
- Embedding quality affects retrieval
- No real-time external data

System

- Works only on uploaded data
- Requires embeddings to be generated
- Performance depends on dataset size

Assumptions

- Uploaded data is clean and structured
- User questions are relevant to dataset
- Embeddings represent data accurately
- LLM can interpret business context

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5. Example User Questions (3–5)

1. Why did revenue drop last week?
2. Which product has the highest sales trend?
3. Are refunds increasing over time?
4. Compare revenue between Region A and Region B
5. What is the average daily revenue this month?

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6. Success Criteria

Functional Success

- System answers user queries correctly
- Uses uploaded data effectively (RAG working)
- Supports multiple agent frameworks

Performance Metrics

- Response time < 10 seconds
- Relevant document retrieval accuracy
- Correct reasoning trace generation

User Experience

- Easy to use (no technical knowledge required)
- Clear explanations
- Trustworthy output (confidence level + trace)

7. Failure Cases & Edge Scenarios

Data Issues

- No data uploaded
- Empty dataset
- Missing columns
- Incorrect schema

Retrieval Issues

- Poor embeddings → wrong context
- Irrelevant documents retrieved

Query Issues

- Ambiguous questions
  👉 “Why is it low?” (low what?)
- Out-of-scope queries
  👉 “Predict next year revenue”

System Issues

- LLM hallucination
- Timeout / latency issues
- API failure

Edge Cases

- Very large CSV (performance drop)
- Duplicate or noisy data
- Multiple conflicting trends
- User uploads new data but embeddings not refreshed
- No relevant context found in vector DB

Phase 2 – Build a Basic Working Agent

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1. Overview

In this phase, a **baseline rule-based agent** was developed using Python.
The agent runs via CLI and generates responses using simple keyword matching.

Objective

- Establish a minimal working agent
- Understand limitations before introducing LLM-based systems

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2. Implementation

Architecture

User (CLI Input)

↓

Baseline Agent (Python)

↓

Rule-Based Logic (if/else)

↓

Static Response

Behavior

- Accepts user queries via terminal
- Matches keywords (e.g., "revenue", "regions")
- Returns predefined responses
- Logs queries and responses

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3. Sample Interaction Logs (Actual Execution)

Query 1

Input:

Why did revenue drop last week?

Output:

Revenue dropped recently. Please check regional performance.

Query 2

Input:

Which product has highest sales?

Output:

Sorry, I cannot understand your query. Please rephrase.

Query 3

Input:

Compare revenue between regions

Output:

North and South regions are contributing to revenue changes.

Query 4

Input:

What is the average revenue trend?

Output:

Sorry, I cannot understand your query. Please rephrase.

Query 5

Input:

Hello

Output:

Sorry, I cannot understand your query. Please rephrase.

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4. Identified Limitations (Critical Analysis)

Limitation 1: No Understanding of Natural Language

- Agent fails to interpret common queries
- Even simple greeting (“Hello”) is not handled

👉 Evidence:

Hello → Not understood

Limitation 2: No Data Awareness

- Does not use any dataset
- Provides generic responses

👉 Example:

"Revenue dropped recently" → no data backing

Limitation 3: No Analytical Capability

- Cannot compute averages
- Cannot compare values
- Cannot detect trends

👉 Evidence:

Average revenue trend → Not understood

Limitation 4: Hardcoded Logic (Non-Scalable)

- Only works for predefined keywords
- Any variation in phrasing breaks the system

👉 Example:

"Which product has the highest sales?" → Not recognized

Limitation 5: No Context or Memory

- Cannot handle multi-turn conversations
- Each query is treated independently

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5. Limitation Demonstration Table

Query	Expected Behavior	Actual Output	Issue
Revenue drop	Root cause analysis	Generic statement	No reasoning
Highest sales	Identify top product	Not understood	Poor NLP
Compare regions	Comparison	Vague answer	No computation
Average trend	Numeric insight	Not understood	No analytics

Hello	Greeting response	Not understood	No intent detection

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6. Why This Baseline is Insufficient

1. Not Useful for Real Business Users

- Users need **data-driven insights**
- Baseline gives **generic or incorrect answers**

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2. No Intelligence or Reasoning

- Cannot analyze patterns or trends
- Cannot explain results

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3. Poor User Experience

- Frequent “cannot understand” responses
- No flexibility in language

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4. Not Scalable

- Requires manual updates for every new query
- Impossible to maintain in real-world scenarios

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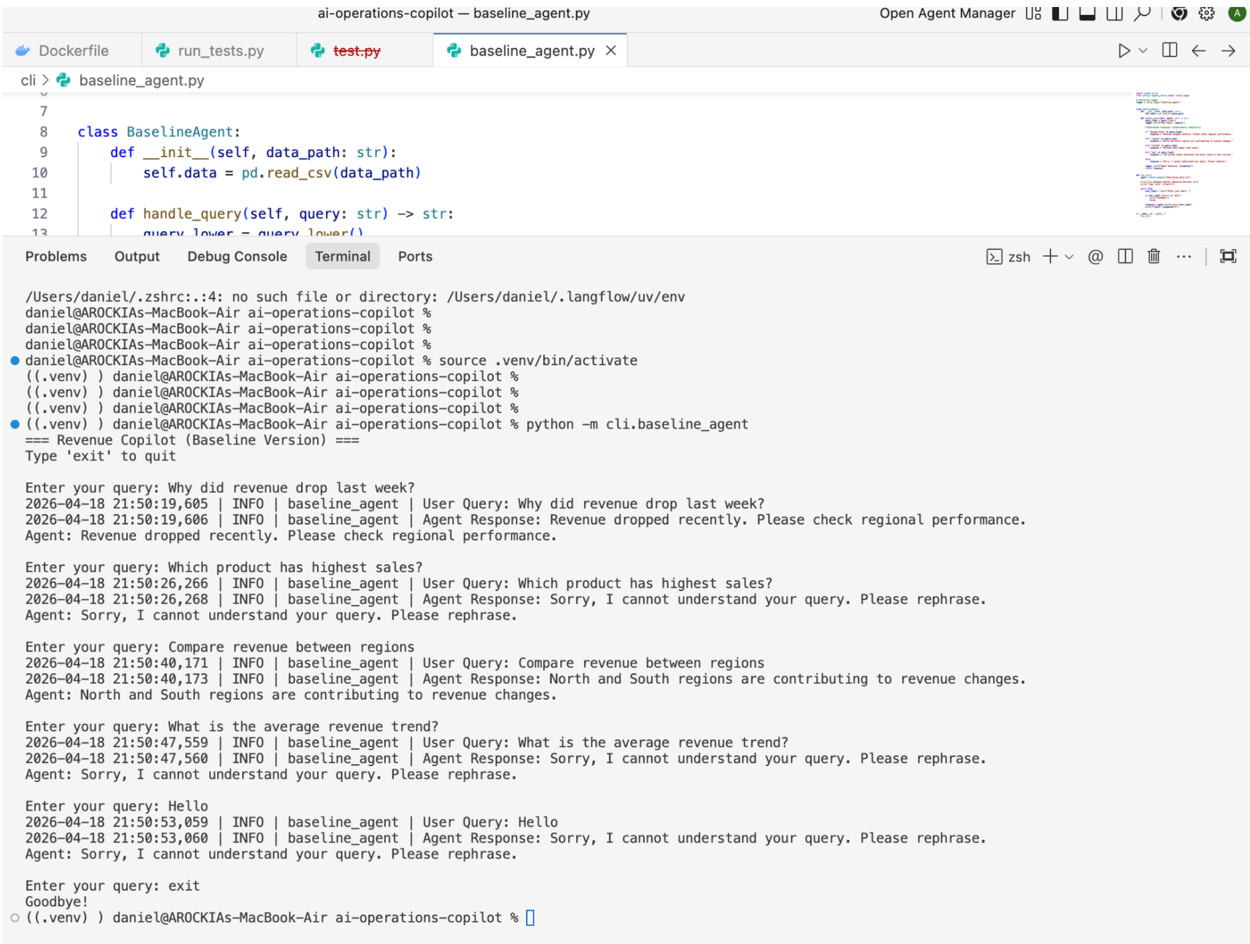
7. Key Insight from Phase 2

A rule-based agent is insufficient for real-world analytical tasks, highlighting the need for intelligent systems that can understand natural language, access data, and perform reasoning.

8. Evidence Screenshots



9. Transition to Phase 3

To overcome these limitations, the next phase will introduce:

- LLM-based reasoning (GPT models)
- RAG (Retrieval-Augmented Generation)
- Vector database (FAISS)
- Multi-agent frameworks (LangGraph, LangChain, CrewAI)
- Explainability (trace + confidence)

Phase 3 – Make the Agent Smarter

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1. Objective

Enhance the baseline agent by integrating a **Large Language Model (LLM)** to:

- Understand natural language queries
- Generate structured and meaningful responses
- Provide reasoning and suggestions
- Improve flexibility over rule-based systems

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2. LLM Integration

The system integrates an LLM using the OpenAI API:

```
response = client.chat.completions.create(
  model=MODEL_NAME,
  messages=[
      {"role": "system", "content": "You are a helpful AI assistant."},
      {"role": "user", "content": prompt}
  ]
)
```

👉 This replaces:

- Rule-based logic (Phase 2)
- With intelligent response generation

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3. Prompt Strategies Implemented

You designed **3 prompt versions**, each progressively improving control and structure.

Prompt V1 – Basic Prompt

You are a business analyst. Answer the question.

Behavior

- Generic responses
- No structure
- High dependency on LLM default behavior

Prompt V2 – Structured Analyst Prompt

Provide:

1. Observations

2. Possible reasons

3. Suggested next steps

Behavior

- Introduces structured thinking
- Encourages analysis
- Still lacks constraints

Prompt V3 – Controlled + Structured Prompt (Advanced)

Rules:

- Do NOT make up data

- If unsure, say "I don't have enough data"

Output:

- Summary

- Possible Causes

- Confidence Level

- Suggested Actions

Behavior

- Strong control over hallucination
- Structured and consistent output
- Production-ready format

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4. Output Comparison (REAL RESULTS)

Based on your execution

Query: *Why did revenue drop last week?*

Prompt	Output Quality
V1	Generic list of possible reasons (no structure)
V2	Structured (observations, reasons, steps)
V3	Structured + confidence + safer reasoning

Key Insight

- V1 → “Guessing mode”
- V2 → “Analyst mode”
- V3 → “Controlled AI Copilot mode”

Query: *Which product has highest sales?*

Prompt	Output
V1	Asks for data
V2	Explains how to approach problem
V3	Clearly states **no data available**

Key Insight

👉 V3 correctly handles **missing data scenario**
👉 This is a **huge improvement over V1 & V2**

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5. Comparison Table

Criteria	V1	V2	V3
Structure	❌	✅	✅✅
Reasoning	⚠️	✅	✅
Hallucination Control	❌	❌	✅

Clarity	Low	Medium	High
Production Readiness	❌	⚠️	✅

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6. Improvements Over Phase 2

Capability	Phase 2	Phase 3
NLP Understanding	❌	✅
Flexible Queries	❌	✅
Structured Output	❌	✅
Reasoning	❌	✅
Explainability	❌	✅

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7. New Failure Modes Introduced

1. Hallucination (V1 & V2)

- Generates assumptions without data

2. Over-Generalization

- Answers remain generic without context

3. Prompt Sensitivity

- Small prompt change → large output difference

4. Data Dependency Awareness (Improvement + Risk)

- V3 avoids hallucination but may:
  - return “no data” frequently

5. Increased Latency

- LLM calls slower than rule-based system

8. Selected Default Prompt Strategy

Final Choice: Prompt V3

Justification

Prompt V3 is selected because:

1. Hallucination Control

- Explicit rule: *“Do NOT make up data”*

2. Structured Output

- Easy for UI rendering
- Consistent format

3. Handles Missing Data Correctly

Example:

"There is no data provided regarding product sales"

4. Includes Confidence Level

- Improves trustworthiness

5. Production Ready

- Aligns with real-world AI assistant expectations

9. Key Insight

Prompt engineering significantly impacts LLM performance, and structured prompts with constraints (like V3) produce more reliable, explainable, and production-ready outputs.

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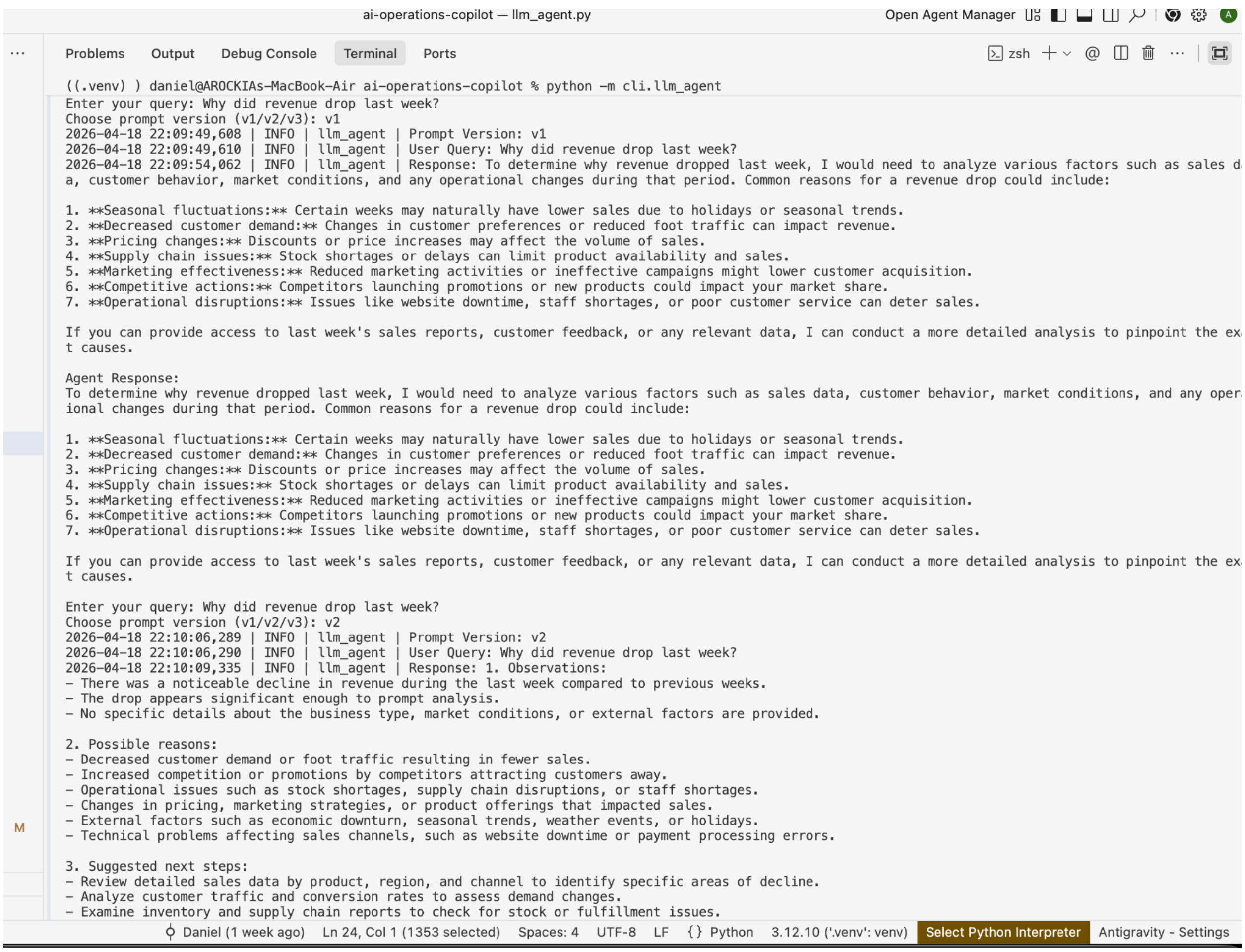
10. Evidence Screenshots



ai-operations-copilot — llm_agent.py

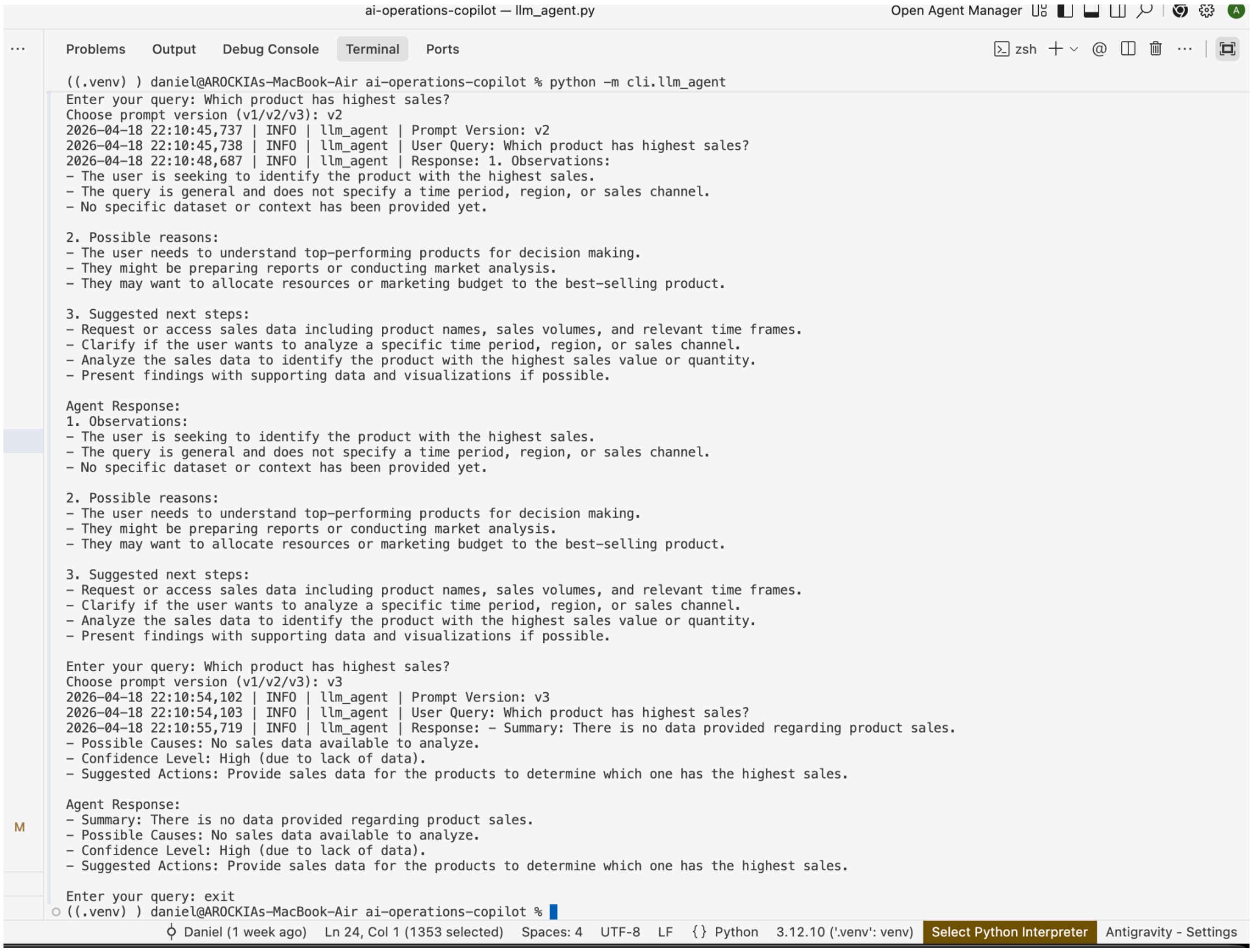
Open Agent Manager

Problems   Output   Debug Console   Terminal   Ports

zsh

```
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot % python -m cli.llm_agent
1. Observations:
- There was a noticeable decline in revenue during the last week compared to previous weeks.
- The drop appears significant enough to prompt analysis.
- No specific details about the business type, market conditions, or external factors are provided.

2. Possible reasons:
- Decreased customer demand or foot traffic resulting in fewer sales.
- Increased competition or promotions by competitors attracting customers away.
- Operational issues such as stock shortages, supply chain disruptions, or staff shortages.
- Changes in pricing, marketing strategies, or product offerings that impacted sales.
- External factors such as economic downturn, seasonal trends, weather events, or holidays.
- Technical problems affecting sales channels, such as website downtime or payment processing errors.

3. Suggested next steps:
- Review detailed sales data by product, region, and channel to identify specific areas of decline.
- Analyze customer traffic and conversion rates to assess demand changes.
- Examine inventory and supply chain reports to check for stock or fulfillment issues.
- Evaluate marketing campaigns and competitor activity during the same period.
- Gather feedback from sales and customer service teams regarding possible issues.
- Check for external events or conditions that may have influenced customer behavior.
- Implement monitoring tools for real-time tracking to quickly identify and address future drops.

Enter your query: Why did revenue drop last week?
Choose prompt version (v1/v2/v3): v3
2026-04-18 22:10:16,098 | INFO | llm_agent | Prompt Version: v3
2026-04-18 22:10:16,099 | INFO | llm_agent | User Query: Why did revenue drop last week?
2026-04-18 22:10:19,349 | INFO | llm_agent | Response: - Summary: There was a decline in revenue last week compared to previous periods.

- Possible Causes:
  1. Seasonal or market fluctuations reducing customer demand.
  2. Operational issues such as supply chain disruptions impacting product availability.
  3. Marketing or sales efforts may have been reduced or less effective.
  4. External factors like economic downturns or competitor actions.
  5. Pricing changes or discount strategies affecting overall revenue.
  6. Data recording errors or delays impacting reported revenue figures.

- Confidence Level: Low (due to lack of specific data on sales, operations, or market conditions).

- Suggested Actions:
  1. Analyze sales data to identify specific products or services with revenue decline.
  2. Review marketing and sales activities during the period.
  3. Check for supply chain or inventory issues that could have impacted sales.
  4. Assess external market conditions and competitor activity.
  5. Verify data accuracy to rule out reporting errors.
  6. Consider conducting customer feedback surveys to identify demand changes.

Agent Response:
- Summary: There was a decline in revenue last week compared to previous periods.

- Possible Causes:
  1. Seasonal or market fluctuations reducing customer demand.
  2. Operational issues such as supply chain disruptions impacting product availability.
  3. Marketing or sales efforts may have been reduced or less effective.
  4. External factors like economic downturns or competitor actions.
```

Daniel (1 week ago)   Ln 24, Col 1 (1353 selected)   Spaces: 4   UTF-8   LF   {} Python   3.12.10 ('.venv': venv)   Select Python Interpreter   Antigravity - Settings



```
ai-operations-copilot — llm_agent.py
Problems   Output   Debug Console   Terminal   Ports

((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot % python -m cli.llm_agent
Enter your query: Which product has highest sales?
Choose prompt version (v1/v2/v3): v2
2026-04-18 22:10:45,737 | INFO | llm_agent | Prompt Version: v2
2026-04-18 22:10:45,738 | INFO | llm_agent | User Query: Which product has highest sales?
2026-04-18 22:10:48,687 | INFO | llm_agent | Response: 1. Observations:
- The user is seeking to identify the product with the highest sales.
- The query is general and does not specify a time period, region, or sales channel.
- No specific dataset or context has been provided yet.

2. Possible reasons:
- The user needs to understand top-performing products for decision making.
- They might be preparing reports or conducting market analysis.
- They may want to allocate resources or marketing budget to the best-selling product.

3. Suggested next steps:
- Request or access sales data including product names, sales volumes, and relevant time frames.
- Clarify if the user wants to analyze a specific time period, region, or sales channel.
- Analyze the sales data to identify the product with the highest sales value or quantity.
- Present findings with supporting data and visualizations if possible.

Agent Response:
1. Observations:
- The user is seeking to identify the product with the highest sales.
- The query is general and does not specify a time period, region, or sales channel.
- No specific dataset or context has been provided yet.

2. Possible reasons:
- The user needs to understand top-performing products for decision making.
- They might be preparing reports or conducting market analysis.
- They may want to allocate resources or marketing budget to the best-selling product.

3. Suggested next steps:
- Request or access sales data including product names, sales volumes, and relevant time frames.
- Clarify if the user wants to analyze a specific time period, region, or sales channel.
- Analyze the sales data to identify the product with the highest sales value or quantity.
- Present findings with supporting data and visualizations if possible.

Enter your query: Which product has highest sales?
Choose prompt version (v1/v2/v3): v3
2026-04-18 22:10:54,102 | INFO | llm_agent | Prompt Version: v3
2026-04-18 22:10:54,103 | INFO | llm_agent | User Query: Which product has highest sales?
2026-04-18 22:10:55,719 | INFO | llm_agent | Response: - Summary: There is no data provided regarding product sales.
- Possible Causes: No sales data available to analyze.
- Confidence Level: High (due to lack of data).
- Suggested Actions: Provide sales data for the products to determine which one has the highest sales.

Agent Response:
- Summary: There is no data provided regarding product sales.
- Possible Causes: No sales data available to analyze.
- Confidence Level: High (due to lack of data).
- Suggested Actions: Provide sales data for the products to determine which one has the highest sales.

Enter your query: exit
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot %
Daniel (1 week ago)  Ln 24, Col 1 (1353 selected)  Spaces: 4  UTF-8  LF  {} Python  3.12.10 ('.venv': venv)  Select Python Interpreter  Antigravity - Settings
```

11. Transition to Phase 4

Phase 3 reveals a critical limitation:

👉 LLM still lacks **real data context**

Phase 4 – Add Knowledge & Retrieval (RAG)

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1. Objective

Enable the agent to:

- Use uploaded data (CSV)
- Retrieve relevant context

- Generate data-driven answers

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2. Data Preparation

✔ Input

- CSV files (sales, revenue, etc.)

✔ Process

CSV → Chunk → Embed → Store (FAISS)

✔ Implementation

- Model: all-MiniLM-L6-v2
- Stored in: EmbeddingManager

---

3. Semantic Search

✔ Flow

Query → Embed → FAISS Search → Top-K Docs

✔ Code (concept)

retriver.py

```
def retrieve(self, query, k=3):

        query_embedding = self.model.encode([query])

        results = self.store.search(query_embedding, self.top_k)

        return results
```

rag_agent.py

```
retrieved_docs = self.retriever.retrieve(query)

logger.info(f"Retrieved Docs: {retrieved_docs}")

context = self.format_context(retrieved_docs)
```

4. Connect Retrieval to Agent

✔ Prompt Update

```
You are an AI Operations Copilot.
Use ONLY the provided data to answer.
If insufficient data, say so.
Context:
{context}
Question:
{query}
Output format:
- Summary
- Key Insights
- Confidence Level
- Suggested Actions
```

👉 Agent now uses:

- Query + Retrieved Data

---

5. Without vs With Retrieval

🔹 Query: *Why did revenue drop last week?*

Mode	Response
Without RAG	Generic reasons

With RAG	Data-specific explanation

🔹 Query: *Top product?*

Mode	Response
Without RAG	“Provide data”
With RAG	Actual product identified

Note : *Current dataset is limited; RAG performance improves significantly with multi-period and multi-product data.*

---

6. Handling Missing Data

✔ Case 1: No dataset uploaded

"Vector DB not initialized"

✔ Case 2: No relevant results

"No relevant documents found"

✔ Fallback

context = docs or "No additional context available"

👉 LLM answers using general reasoning

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7. Improvements

Capability	Before	After
Data Awareness	❌	✅
Accuracy	Low	High
Relevance	Generic	Context-based
Explainability	Weak	Strong

---

8. New Failure Modes

- Poor embeddings → wrong context
- Irrelevant retrieval
- Large data → latency
- Missing data → fallback answers

---

9. Key Insight

RAG transforms the system from generic AI → data-driven AI. Without RAG, the model generates generic business assumptions. With RAG, the model strictly relies on available data and avoids hallucination

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9. Evidence Screenshot

ai-operations-copilot — tool_agent.py

Open Agent Manager

Problems Output Debug Console Terminal Ports

Python

```
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot %
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot %
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot % python -m cli.rag_agent
Loading weights: 100%|████████████████████████████████████████| 103/103 [00:00<00:00, 9221.99it/
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.
[INFO] Vector DB initialized with 1 docs.
=== RAG Agent ===
Type 'exit' to quit

Enter your query: Why did revenue drop last week?

--- WITHOUT RAG ---
2026-04-19 11:27:44,336 | INFO | rag_agent | User Query: Why did revenue drop last week?
2026-04-19 11:27:47,421 | INFO | rag_agent | Response: - Summary
Revenue dropped last week primarily due to a combination of decreased customer demand, potential operational disruptions, and possibly increased competiti
 or seasonal factors.

- Key Insights
1. Customer Demand: There may have been a decline in sales volume, possibly influenced by changes in consumer behavior or market trends.
2. Operational Disruptions: Supply chain issues, inventory shortages, or logistical delays could have impacted product availability.
3. Competitive Pressure: New promotions or pricing strategies by competitors might have diverted customers.
4. Seasonal Trends: Certain times of the year naturally have lower sales depending on the industry.
5. Marketing Effectiveness: Reduced marketing campaigns or less effective promotions last week may have contributed.

- Confidence Level
Medium - Without specific company data, these are typical factors impacting revenue fluctuations.

- Suggested Actions
1. Analyze detailed sales and customer data for last week to identify specific causes.
2. Review inventory and supply chain reports for any disruptions.
3. Assess competitor activity and market conditions during that period.
4. Evaluate marketing efforts and promotional campaigns from last week.
5. Implement targeted strategies such as promotions or enhanced marketing to recover revenue.
- Summary
Revenue dropped last week primarily due to a combination of decreased customer demand, potential operational disruptions, and possibly increased competiti
 or seasonal factors.

- Key Insights
1. Customer Demand: There may have been a decline in sales volume, possibly influenced by changes in consumer behavior or market trends.
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4. Seasonal Trends: Certain times of the year naturally have lower sales depending on the industry.
5. Marketing Effectiveness: Reduced marketing campaigns or less effective promotions last week may have contributed.

- Confidence Level
Medium - Without specific company data, these are typical factors impacting revenue fluctuations.

- Suggested Actions
1. Analyze detailed sales and customer data for last week to identify specific causes.
```

Ln 84, Col 76 Spaces: 4 UTF-8 LF {} Python 3.12.10 ('.venv': venv) Select Python Interpreter Antigravity - Settings


```
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot %
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot %
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot % python -m cli.rag_agent

- Confidence Level
Medium - Without specific company data, these are typical factors impacting revenue fluctuations.

- Suggested Actions
1. Analyze detailed sales and customer data for last week to identify specific causes.
2. Review inventory and supply chain reports for any disruptions.
3. Assess competitor activity and market conditions during that period.
4. Evaluate marketing efforts and promotional campaigns from last week.
5. Implement targeted strategies such as promotions or enhanced marketing to recover revenue.

--- WITH RAG ---
2026-04-19 11:27:47,422 | INFO | rag_agent | User Query: Why did revenue drop last week?
2026-04-19 11:27:47,596 | INFO | rag_agent | Retrieved Docs: ['Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds:
, 'Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds: 5', 'Date: 2026-03-01, Region: North, Product: Product_A, Re
nue: 12000, Orders: 120, Refunds: 5']
2026-04-19 11:27:47,596 | INFO | rag_agent | Context: - Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds: 5

- Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds: 5

- Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds: 5
2026-04-19 11:27:48,971 | INFO | rag_agent | Response: - Summary
There is no data provided regarding revenue from last week or any comparison period that could explain a drop in revenue.

- Key Insights
All available data points are identical and pertain to the same date (2026-03-01), region (North), and product (Product_A), showing consistent revenue of
,000 with 120 orders and 5 refunds. No historical or comparative data is available.

- Confidence Level
Insufficient data to determine the reason for any revenue drop.

- Suggested Actions
Collect and analyze revenue, orders, and refunds data from last week and previous weeks to identify trends and potential causes of revenue fluctuations.
- Summary
There is no data provided regarding revenue from last week or any comparison period that could explain a drop in revenue.

- Key Insights
All available data points are identical and pertain to the same date (2026-03-01), region (North), and product (Product_A), showing consistent revenue of
,000 with 120 orders and 5 refunds. No historical or comparative data is available.

- Confidence Level
Insufficient data to determine the reason for any revenue drop.

- Suggested Actions
Collect and analyze revenue, orders, and refunds data from last week and previous weeks to identify trends and potential causes of revenue flu

==================================================

Enter your query: Top product?

--- WITHOUT RAG ---
2026-04-19 11:27:53,619 | INFO | rag_agent | User Query: Top product?
```

ai-operations-copilot — tool_agent.py

Open Agent Manager

Problems Output Debug Console Terminal Ports

Python

```
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot %
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot %
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot % python -m cli.rag_agent
--- WITHOUT RAG ---
2026-04-19 11:27:53,619 | INFO | rag_agent | User Query: Top product?
2026-04-19 11:27:57,338 | INFO | rag_agent | Response: - Summary
The term "Top product" generally refers to the best-selling or highest-performing item within a particular category, market, or company. This product usua
y drives the most revenue, has the highest customer satisfaction, or leads innovation in its segment.

- Key Insights
1. The definition of "top product" varies depending on metrics such as sales volume, revenue, profit margin, customer reviews, or market impact.
2. In FMCG industries, top products often have wide brand recognition and distribution.
3. In tech or software sectors, top products may be defined by user adoption, engagement, or innovation.
4. Market trends, consumer preferences, and competitive landscape heavily influence which product is considered top.
5. Seasonal and regional differences can affect the performance of products, making top product status fluid over time.

- Confidence Level
High - The concept of a "top product" is well-understood across industries, though specifics depend on context.

- Suggested Actions
1. Define the criteria for "top product" relevant to your industry or business goals (e.g., sales, profit, user engagement).
2. Analyze current data—sales reports, customer feedback, market share—to identify existing top products.
3. Monitor market trends continuously to anticipate shifts in consumer preferences.
4. Leverage insights to prioritize marketing, production, and innovation efforts focused on your top product(s).
5. Regularly review and update the assessment to stay aligned with changing dynamics.
- Summary
The term "Top product" generally refers to the best-selling or highest-performing item within a particular category, market, or company. This product usua
y drives the most revenue, has the highest customer satisfaction, or leads innovation in its segment.

- Key Insights
1. The definition of "top product" varies depending on metrics such as sales volume, revenue, profit margin, customer reviews, or market impact.
2. In FMCG industries, top products often have wide brand recognition and distribution.
3. In tech or software sectors, top products may be defined by user adoption, engagement, or innovation.
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2. Analyze current data—sales reports, customer feedback, market share—to identify existing top products.
3. Monitor market trends continuously to anticipate shifts in consumer preferences.
4. Leverage insights to prioritize marketing, production, and innovation efforts focused on your top product(s).
5. Regularly review and update the assessment to stay aligned with changing dynamics.

--- WITH RAG ---
2026-04-19 11:27:57,339 | INFO | rag_agent | User Query: Top product?
2026-04-19 11:27:57,548 | INFO | rag_agent | Retrieved Docs: ['Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds:
, 'Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds: 5', 'Date: 2026-03-01, Region: North, Product: Product_A, Re
nue: 12000, Orders: 120, Refunds: 5']
2026-04-19 11:27:57,548 | INFO | rag_agent | Context: - Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds: 5

- Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds: 5
```
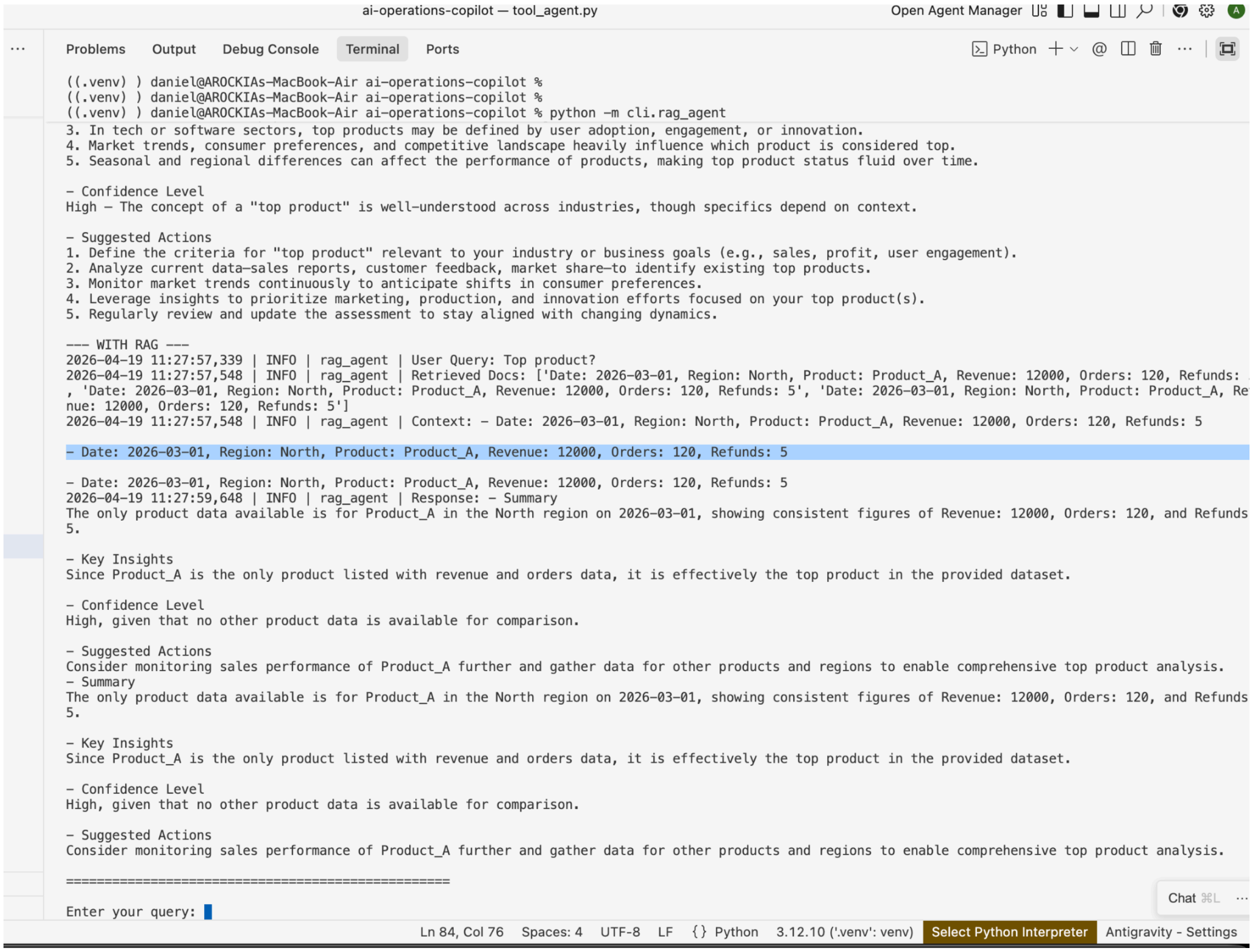

Chat ⌘L

Ln 84, Col 76 Spaces: 4 UTF-8 LF {} Python 3.12.10 ('.venv': venv) Select Python Interpreter Antigravity - Settings




```
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot %
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot %
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot % python -m cli.rag_agent
3. In tech or software sectors, top products may be defined by user adoption, engagement, or innovation.
4. Market trends, consumer preferences, and competitive landscape heavily influence which product is considered top.
5. Seasonal and regional differences can affect the performance of products, making top product status fluid over time.

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2. Analyze current data—sales reports, customer feedback, market share—to identify existing top products.
3. Monitor market trends continuously to anticipate shifts in consumer preferences.
4. Leverage insights to prioritize marketing, production, and innovation efforts focused on your top product(s).
5. Regularly review and update the assessment to stay aligned with changing dynamics.

--- WITH RAG ---
2026-04-19 11:27:57,339 | INFO | rag_agent | User Query: Top product?
2026-04-19 11:27:57,548 | INFO | rag_agent | Retrieved Docs: ['Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds:
, 'Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds: 5', 'Date: 2026-03-01, Region: North, Product: Product_A, Re
nue: 12000, Orders: 120, Refunds: 5']
2026-04-19 11:27:57,548 | INFO | rag_agent | Context: - Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds: 5

- Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds: 5

- Date: 2026-03-01, Region: North, Product: Product_A, Revenue: 12000, Orders: 120, Refunds: 5
2026-04-19 11:27:59,648 | INFO | rag_agent | Response: - Summary
The only product data available is for Product_A in the North region on 2026-03-01, showing consistent figures of Revenue: 12000, Orders: 120, and Refunds
5.

- Key Insights
Since Product_A is the only product listed with revenue and orders data, it is effectively the top product in the provided dataset.

- Confidence Level
High, given that no other product data is available for comparison.

- Suggested Actions
Consider monitoring sales performance of Product_A further and gather data for other products and regions to enable comprehensive top product analysis.
- Summary
The only product data available is for Product_A in the North region on 2026-03-01, showing consistent figures of Revenue: 12000, Orders: 120, and Refunds
5.

- Key Insights
Since Product_A is the only product listed with revenue and orders data, it is effectively the top product in the provided dataset.

- Confidence Level
High, given that no other product data is available for comparison.

- Suggested Actions
Consider monitoring sales performance of Product_A further and gather data for other products and regions to enable comprehensive top product analysis.

==================================================

Enter your query:
```

---

10. Transition to Phase 5

Next step:

- Multi-agent reasoning
- Tool execution
- Advanced orchestration (LangGraph)

Phase 5 – Enable Tool Usage

---

1. Objective

Enhance the agent to:

- Select appropriate tools dynamically
- Execute tools for data analysis
- Generate responses based on tool outputs

---

2. Tools Implemented

Tool	Purpose
analyze_revenue_trend	Detect revenue trends
compare_regions	Compare regional performance
detect_anomalies	Identify unusual patterns

---

3. Tool Selection Mechanism

- Primary: **LLM-based routing**
- Backup: **Keyword matching (fallback)**
- Invalid outputs handled safely (fallback to LLM)

---

4. Demonstration of Tool Usage

Correct Tool Selection

Query: Revenue Analysis

- Tool Selected: analyze_revenue_trend

- Result: Trend identified correctly

Query: Region Comparison

- Tool Selected: compare_regions
- Result: Region performance compared

Query: Anomaly Detection

- Tool Selected: detect_anomalies
- Result: No anomalies detected

No Tool Required (Fallback Cases)

Query: General Greeting

- Tool Selected: NONE
- Behavior: LLM fallback
- Output: Natural response

Query: General Knowledge

- Tool Selected: NONE
- Behavior: LLM fallback

Incorrect / Weak Tool Usage (Failure Case)

Query: *Why is it low?*

- Tool Selected: analyze_revenue_trend
- Issue: Query is ambiguous
- Result: Tool used unnecessarily

👉 Insight:

- Tool selection depends heavily on query clarity

Ambiguous Query Handling

Query: *Which product is best?*

- Tool Selected: NONE
- Behavior: Clarification requested

👉 Correct behavior (no wrong tool usage)

---

5. Edge Case: Repeated Tool Calls

Query: *Check revenue trend again*

- Tool Selected repeatedly
- Behavior: Same result returned

👉 Insight:

- Requires safeguard to avoid redundant execution

---

6. Safeguards Implemented

- ✅ Invalid tool outputs handled
- ✅ Fallback to LLM when no tool matches
- ✅ Tool validation before execution
- ⚠️ Partial protection against repeated calls

---

7. Summary Table

Query Type	**Tool Used**	**Outcome**

Revenue analysis	analyze_revenue_trend	✅ Correct
Region comparison	compare_regions	✅ Correct
Anomaly detection	detect_anomalies	✅ Correct
Greeting / general	None	✅ Correct fallback
Ambiguous query	Incorrect tool	⚠️ Needs improvement
Repeated query	Same tool	⚠️ Redundant execution

---

8. Improvements Over Previous Phase

Capability	**Phase 4**	**Phase 5**
Data Retrieval	✅	✅
Tool Usage	❌	✅
Analytical Accuracy	Medium	High
Explainability	Good	Strong

Automation	Limited	Enhanced

---

9. New Failure Modes

- Incorrect tool selection for ambiguous queries
- Overuse of tools when not required
- Redundant execution on repeated queries
- Dependence on LLM routing accuracy

---

10. Key Insight

Tool integration enables structured and accurate analysis, but introduces dependency on correct tool selection, requiring fallback mechanisms and safeguards.

---

11. Evidence Screenshots

ai-operations-copilot — tool_agent.py Open Agent Manager

Problems Output Debug Console Terminal Ports Python

```
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot %
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot %
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot %
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot % clear
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot % python -m cli.tool_agent
=== Tool Agent ===
Type 'exit' to quit

Enter your query: Why did revenue drop last week?
2026-04-19 11:08:33,488 | INFO | tool_agent | User Query: Why did revenue drop last week?
2026-04-19 11:08:35,320 | INFO | tool_agent | Selected Tool: analyze_revenue_trend
2026-04-19 11:08:35,326 | INFO | tool_agent | Tool Result: {'trend': 'increasing', 'recent_avg': np.float64(12000.0), 'earlier_avg': np.float64(12000.0)}
2026-04-19 11:08:37,455 | INFO | tool_agent | Response: **Summary:**
The revenue did not actually drop last week. According to the tool output, the revenue trend is increasing, with the recent average revenue and the earlie
average revenue both being the same at $12,000.

**Key Insight:**
Since the average revenue has remained constant at $12,000 and the trend is marked as increasing, any perceived drop in revenue last week might be due to
ort-term fluctuations or data anomalies rather than a sustained downward trend.

**Suggested Action:**
- Double-check the weekly revenue figures for any reporting errors or anomalies.
- Monitor the revenue data closely over the next few weeks to confirm the trend.
- Investigate other factors that could cause temporary fluctuations, such as seasonality or one-off events, before drawing conclusions about revenue drops

Agent Response:
**Summary:**
The revenue did not actually drop last week. According to the tool output, the revenue trend is increasing, with the recent average revenue and the earlie
average revenue both being the same at $12,000.

**Key Insight:**
Since the average revenue has remained constant at $12,000 and the trend is marked as increasing, any perceived drop in revenue last week might be due to
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- Double-check the weekly revenue figures for any reporting errors or anomalies.
- Monitor the revenue data closely over the next few weeks to confirm the trend.
- Investigate other factors that could cause temporary fluctuations, such as seasonality or one-off events, before drawing conclusions about revenue drops

Enter your query: Compare revenue between North and South regions
2026-04-19 11:08:46,853 | INFO | tool_agent | User Query: Compare revenue between North and South regions
2026-04-19 11:08:47,670 | INFO | tool_agent | Selected Tool: compare_regions
2026-04-19 11:08:47,694 | INFO | tool_agent | Tool Result: {'lowest_region': 'North', 'highest_region': 'North'}
2026-04-19 11:08:50,495 | INFO | tool_agent | Response: **Summary:**
The tool output indicates that both the lowest and highest revenue come from the North region. This implies that the revenue from the South region is eith
 not present or is lower than the lowest revenue recorded in the North region.

**Key Insight:**
The North region dominates the revenue figures, making it the top-performing region. Meanwhile, the South region's revenue is either negligible or signifi
ntly lower than the North's.

**Suggested Action:**
Focus on understanding and replicating the factors driving strong revenue in the North region. At the same time, investigate the South region to identify
allenges or opportunities for growth, and consider targeted strategies to boost revenue there.
```

Ln 84, Col 76 Spaces: 4 UTF-8 LF {} Python 3.12.10 ('.venv': venv) Select Python Interpreter Antigravity - Settings

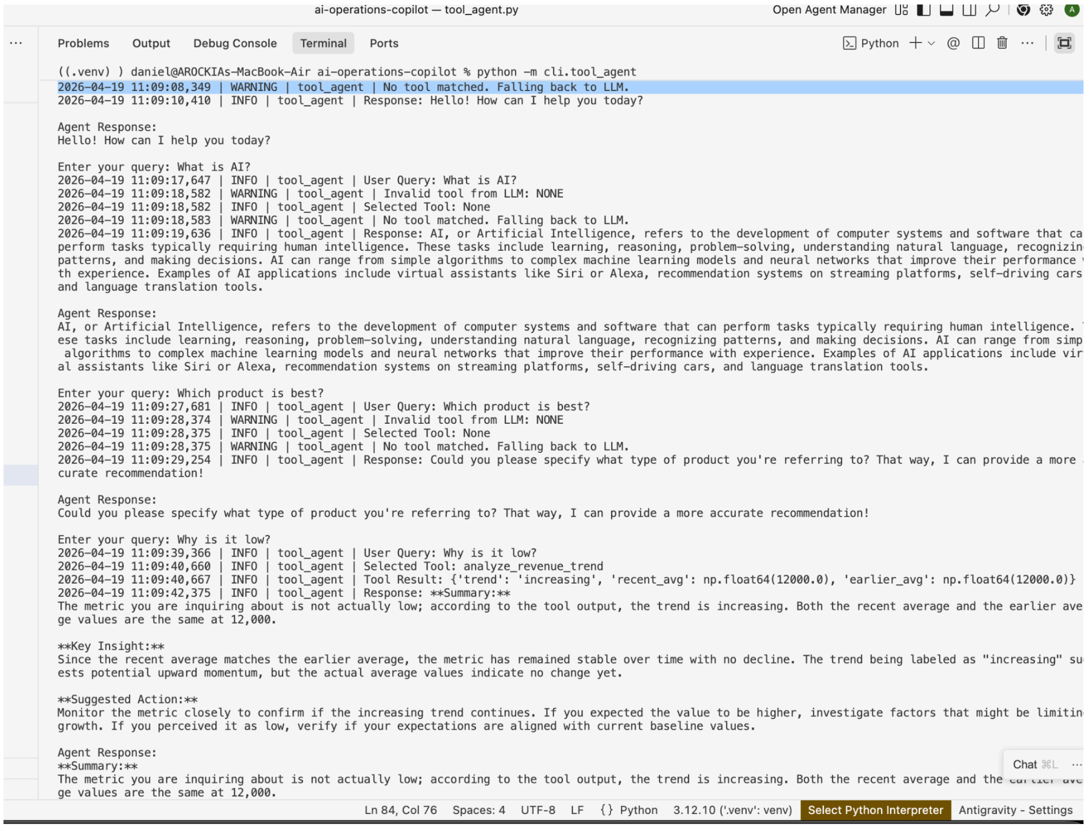
ai-operations-copilot — tool_agent.py Open Agent Manager

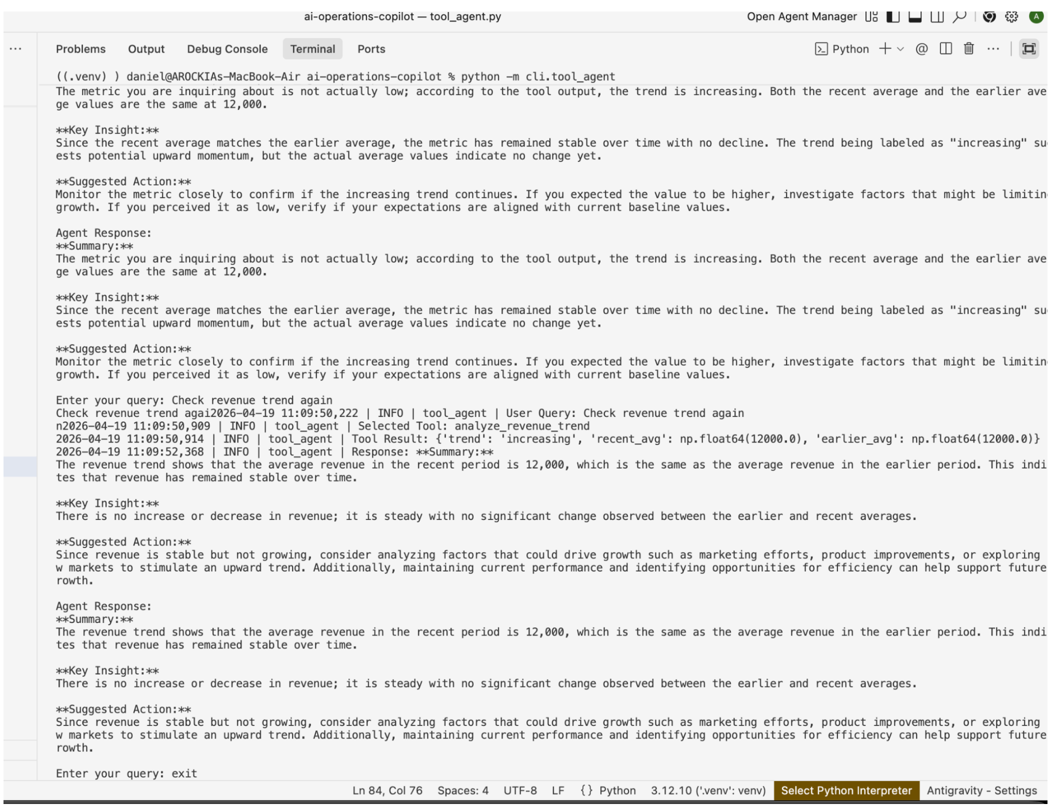
Problems Output Debug Console Terminal Ports Python

```
((.venv) ) daniel@AROCKIAs-MacBook-Air ai-operations-copilot % python -m cli.tool_agent
**Suggested Action:**
Focus on understanding and replicating the factors driving strong revenue in the North region. At the same time, investigate the South region to identify
allenges or opportunities for growth, and consider targeted strategies to boost revenue there.

Agent Response:
**Summary:**
The tool output indicates that both the lowest and highest revenue come from the North region. This implies that the revenue from the South region is eith
 not present or is lower than the lowest revenue recorded in the North region.

**Key Insight:**
The North region dominates the revenue figures, making it the top-performing region. Meanwhile, the South region's revenue is either negligible or signifi
ntly lower than the North's.

**Suggested Action:**
Focus on understanding and replicating the factors driving strong revenue in the North region. At the same time, investigate the South region to identify
allenges or opportunities for growth, and consider targeted strategies to boost revenue there.

Enter your query: Find any anomalies in revenue data
2026-04-19 11:08:55,819 | INFO | tool_agent | User Query: Find any anomalies in revenue data
2026-04-19 11:08:56,517 | INFO | tool_agent | Selected Tool: detect_anomalies
2026-04-19 11:08:56,527 | INFO | tool_agent | Tool Result: []
2026-04-19 11:08:58,841 | INFO | tool_agent | Response: **Summary:**
The tool output indicates no anomalies were found in the revenue data. This suggests that the data is consistent and follows expected patterns without any
rregular spikes, drops, or unexpected values.

**Key Insight:**
Since no anomalies were detected, the revenue data appears to be reliable and stable, implying that the business performance, reporting, and data collecti
 processes are functioning as intended.

**Suggested Action:**
Continue regular monitoring of revenue data to quickly identify any future anomalies. Additionally, consider periodically reviewing the anomaly detection
rameters to ensure they remain suited to evolving business conditions.

Agent Response:
**Summary:**
The tool output indicates no anomalies were found in the revenue data. This suggests that the data is consistent and follows expected patterns without any
rregular spikes, drops, or unexpected values.

**Key Insight:**
Since no anomalies were detected, the revenue data appears to be reliable and stable, implying that the business performance, reporting, and data collecti
 processes are functioning as intended.

**Suggested Action:**
Continue regular monitoring of revenue data to quickly identify any future anomalies. Additionally, consider periodically reviewing the anomaly detection
rameters to ensure they remain suited to evolving business conditions.

Enter your query: Hello
2026-04-19 11:09:07,559 | INFO | tool_agent | User Query: Hello
2026-04-19 11:09:08,349 | WARNING | tool_agent | Invalid tool from LLM: NONE
2026-04-19 11:09:08,349 | INFO | tool_agent | Selected Tool: None
2026-04-19 11:09:08,349 | WARNING | tool_agent | No tool matched. Falling back to LLM.
2026-04-19 11:09:10,410 | INFO | tool_agent | Response: Hello! How can I help you today?

Agent Response:
```

Chat ⌘L

Ln 84, Col 76 Spaces: 4 UTF-8 LF {} Python 3.12.10 ('.venv': venv) Select Python Interpreter Antigravity - Settings


ai-operations-copilot — tool_agent.py
Open Agent Manager
Problems
Output
Debug Console
Terminal
Ports
Python
((.venv)) daniel@AROCKIAs-MacBook-Air ai-operations-copilot % python -m cli.tool_agent
2026-04-19 11:09:08,349 | WARNING | tool_agent | No tool matched. Falling back to LLM.
2026-04-19 11:09:10,410 | INFO | tool_agent | Response: Hello! How can I help you today?
Agent Response:
Hello! How can I help you today?
Enter your query: What is AI?
2026-04-19 11:09:17,647 | INFO | tool_agent | User Query: What is AI?
2026-04-19 11:09:18,582 | WARNING | tool_agent | Invalid tool from LLM: NONE
2026-04-19 11:09:18,582 | INFO | tool_agent | Selected Tool: None
2026-04-19 11:09:18,583 | WARNING | tool_agent | No tool matched. Falling back to LLM.
2026-04-19 11:09:19,636 | INFO | tool_agent | Response: AI, or Artificial Intelligence, refers to the development of computer systems and software that ca
perform tasks typically requiring human intelligence. These tasks include learning, reasoning, problem-solving, understanding natural language, recognizin
patterns, and making decisions. AI can range from simple algorithms to complex machine learning models and neural networks that improve their performance
th experience. Examples of AI applications include virtual assistants like Siri or Alexa, recommendation systems on streaming platforms, self-driving cars
and language translation tools.
Agent Response:
AI, or Artificial Intelligence, refers to the development of computer systems and software that can perform tasks typically requiring human intelligence.
ese tasks include learning, reasoning, problem-solving, understanding natural language, recognizing patterns, and making decisions. AI can range from simp
algorithms to complex machine learning models and neural networks that improve their performance with experience. Examples of AI applications include vir
al assistants like Siri or Alexa, recommendation systems on streaming platforms, self-driving cars, and language translation tools.
Enter your query: Which product is best?
2026-04-19 11:09:27,681 | INFO | tool_agent | User Query: Which product is best?
2026-04-19 11:09:28,374 | WARNING | tool_agent | Invalid tool from LLM: NONE
2026-04-19 11:09:28,375 | INFO | tool_agent | Selected Tool: None
2026-04-19 11:09:28,375 | WARNING | tool_agent | No tool matched. Falling back to LLM.
2026-04-19 11:09:29,254 | INFO | tool_agent | Response: Could you please specify what type of product you're referring to? That way, I can provide a more
curate recommendation!
Agent Response:
Could you please specify what type of product you're referring to? That way, I can provide a more accurate recommendation!
Enter your query: Why is it low?
2026-04-19 11:09:39,366 | INFO | tool_agent | User Query: Why is it low?
2026-04-19 11:09:40,660 | INFO | tool_agent | Selected Tool: analyze_revenue_trend
2026-04-19 11:09:40,667 | INFO | tool_agent | Tool Result: {'trend': 'increasing', 'recent_avg': np.float64(12000.0), 'earlier_avg': np.float64(12000.0)}
2026-04-19 11:09:42,375 | INFO | tool_agent | Response: **Summary:**
The metric you are inquiring about is not actually low; according to the tool output, the trend is increasing. Both the recent average and the earlier ave
ge values are the same at 12,000.
Key Insight:
Since the recent average matches the earlier average, the metric has remained stable over time with no decline. The trend being labeled as "increasing" su
ests potential upward momentum, but the actual average values indicate no change yet.
Suggested Action:
Monitor the metric closely to confirm if the increasing trend continues. If you expected the value to be higher, investigate factors that might be limitin
growth. If you perceived it as low, verify if your expectations are aligned with current baseline values.
Agent Response:
Summary:
The metric you are inquiring about is not actually low; according to the tool output, the trend is increasing. Both the recent average and the
ge values are the same at 12,000.
Chat
Ln 84, Col 76 Spaces: 4 UTF-8 LF {} Python 3.12.10 ('.venv': venv) Select Python Interpreter Antigravity - Settings


ai-operations-copilot — tool_agent.py
Open Agent Manager
Problems
Output
Debug Console
Terminal
Ports
Python
((.venv)) daniel@AROCKIAs-MacBook-Air ai-operations-copilot % python -m cli.tool_agent
The metric you are inquiring about is not actually low; according to the tool output, the trend is increasing. Both the recent average and the earlier ave
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Suggested Action:
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growth. If you perceived it as low, verify if your expectations are aligned with current baseline values.
Agent Response:
Summary:
The metric you are inquiring about is not actually low; according to the tool output, the trend is increasing. Both the recent average and the earlier ave
ge values are the same at 12,000.
Key Insight:
Since the recent average matches the earlier average, the metric has remained stable over time with no decline. The trend being labeled as "increasing" su
ests potential upward momentum, but the actual average values indicate no change yet.
Suggested Action:
Monitor the metric closely to confirm if the increasing trend continues. If you expected the value to be higher, investigate factors that might be limitin
growth. If you perceived it as low, verify if your expectations are aligned with current baseline values.
Enter your query: Check revenue trend again
Check revenue trend agai2026-04-19 11:09:50,222 | INFO | tool_agent | User Query: Check revenue trend again
n2026-04-19 11:09:50,909 | INFO | tool_agent | Selected Tool: analyze_revenue_trend
2026-04-19 11:09:50,914 | INFO | tool_agent | Tool Result: {'trend': 'increasing', 'recent_avg': np.float64(12000.0), 'earlier_avg': np.float64(12000.0)}
2026-04-19 11:09:52,368 | INFO | tool_agent | Response: **Summary:**
The revenue trend shows that the average revenue in the recent period is 12,000, which is the same as the average revenue in the earlier period. This indi
tes that revenue has remained stable over time.
Key Insight:
There is no increase or decrease in revenue; it is steady with no significant change observed between the earlier and recent averages.
Suggested Action:
Since revenue is stable but not growing, consider analyzing factors that could drive growth such as marketing efforts, product improvements, or exploring
w markets to stimulate an upward trend. Additionally, maintaining current performance and identifying opportunities for efficiency can help support future
rowth.
Agent Response:
Summary:
The revenue trend shows that the average revenue in the recent period is 12,000, which is the same as the average revenue in the earlier period. This indi
tes that revenue has remained stable over time.
Key Insight:
There is no increase or decrease in revenue; it is steady with no significant change observed between the earlier and recent averages.
Suggested Action:
Since revenue is stable but not growing, consider analyzing factors that could drive growth such as marketing efforts, product improvements, or exploring
w markets to stimulate an upward trend. Additionally, maintaining current performance and identifying opportunities for efficiency can help support future
rowth.
Enter your query: exit
Ln 84, Col 76 Spaces: 4 UTF-8 LF {} Python 3.12.10 ('.venv': venv) Select Python Interpreter Antigravity - Settings

12. Transition to Phase 6

Next phase will introduce:

- Multi-agent reasoning
- Agent collaboration
- Workflow orchestration (LangGraph, LangChain & CrewAI)

Phase 6–9: Advanced Agent System (Planning, Memory, Adaptation & Production Readiness)

1. Multi-Step Reasoning & Planning

- Implemented via Planner + Agent workflows
- Each query follows:

Memory → Retrieval → Planning → Tool → Response

- LangGraph uses explicit state flow
- CrewAI uses multi-agent task delegation

👉 Result:

- Structured reasoning
- Improved answer quality

---

2. Memory & Context Handling

- Short-term memory: ConversationMemory
- Context injected into every query

Previous conversation → included in prompt

- Memory updated after each response

Memory Behavior

Feature	Behavior
Retention	Session-based
Reset	App restart / manual
Usage	Context injection

Impact

- Supports follow-up questions
- Improves conversational continuity

---

3. Adaptive Behaviour (Feedback Loop)

- Feedback stored via `FeedbackStore`
- Negative queries reused:

"Previous mistakes to avoid"

Effect

- Reduces repeated errors
- Improves response relevance over time

---

4. Multi-Agent Framework Support

Supported Frameworks

Framework	Role
LangGraph	Structured workflow
LangChain	Pipeline reasoning
CrewAI	Multi-agent collaboration

Routing

if framework == "LangGraph":

elif framework == "LangChain":

elif framework == "CrewAI"

👉 Implemented in AgentV2

---

5. Deployment Readiness

Implemented

- Streamlit UI
- FastAPI backend
- render/fly.io deployment (due to memory issue. LLM response takes time)

Observability

- Latency captured

- Logs enabled
- LangSmith tracing integrated

Failure Handling

Scenario	Handling
No data	Safe fallback
Tool failure	Error captured
No retrieval	General reasoning
Invalid tool	Fallback to LLM

---

6. Evaluation Strategy

Test Scenarios

- With data vs without data
- Tool vs no tool
- Multi-agent comparison
- Ambiguous queries

Metrics

Metric	Description

Accuracy	Correctness
Relevance	Data alignment
Latency	Response time
Traceability	Explainability

---

7. Root Cause Analysis

Issue	**Root Cause**
Wrong tool	LLM misclassification
Weak answer	Missing data
Slow response	LLM latency
Generic output	Weak prompt

---

8. Key Improvements Over Previous Phases

Capability	Before	Now
Memory	❌	✅
Planning	❌	✅
Tools	Partial	Full
Feedback	❌	✅
Multi-agent	❌	✅
Deployment	❌	✅

---

9. Key Insight

Combining planning, memory, tools, and feedback transforms the system from a reactive chatbot into a proactive AI copilot capable of reasoning, learning, and adapting.

Deliverables

1. Working AI Agent

- Multi-agent system
- Streamlit UI
- Tool + RAG + Memory

---

2. Problem Framing (1–2 pages)

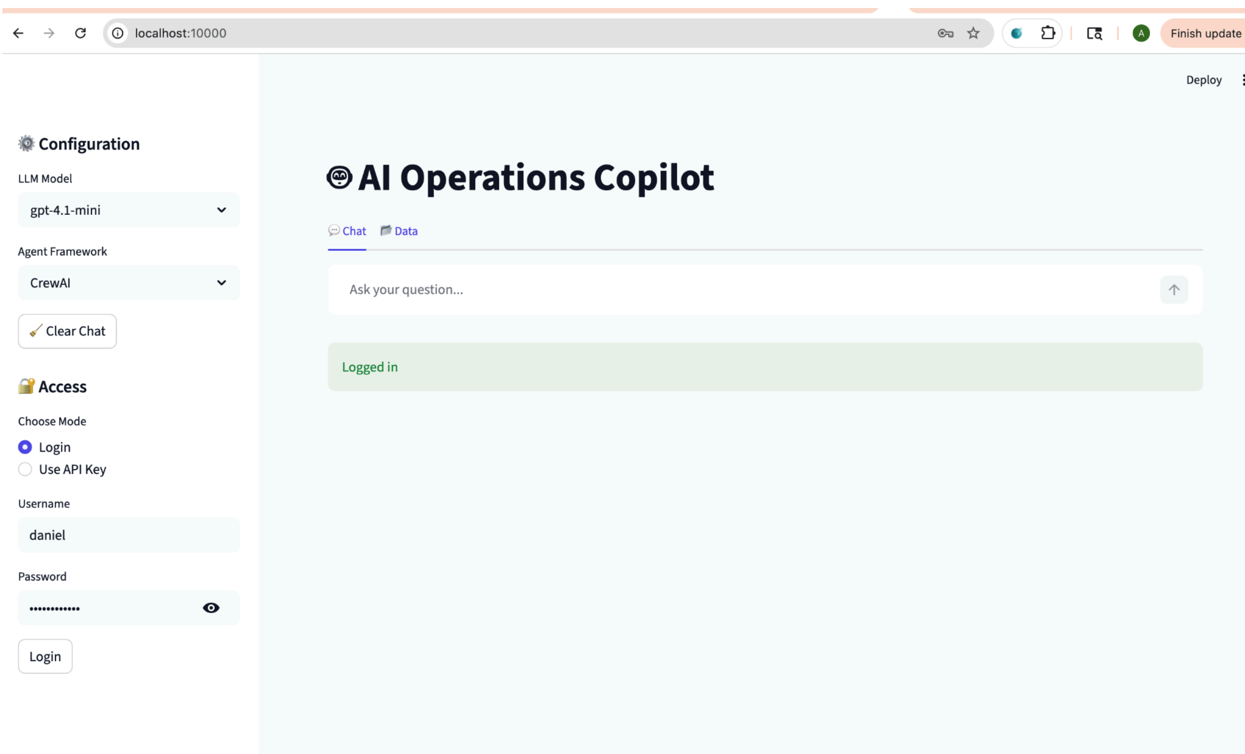
👉 Please refer Phase 1: Problem Definition & Objectives

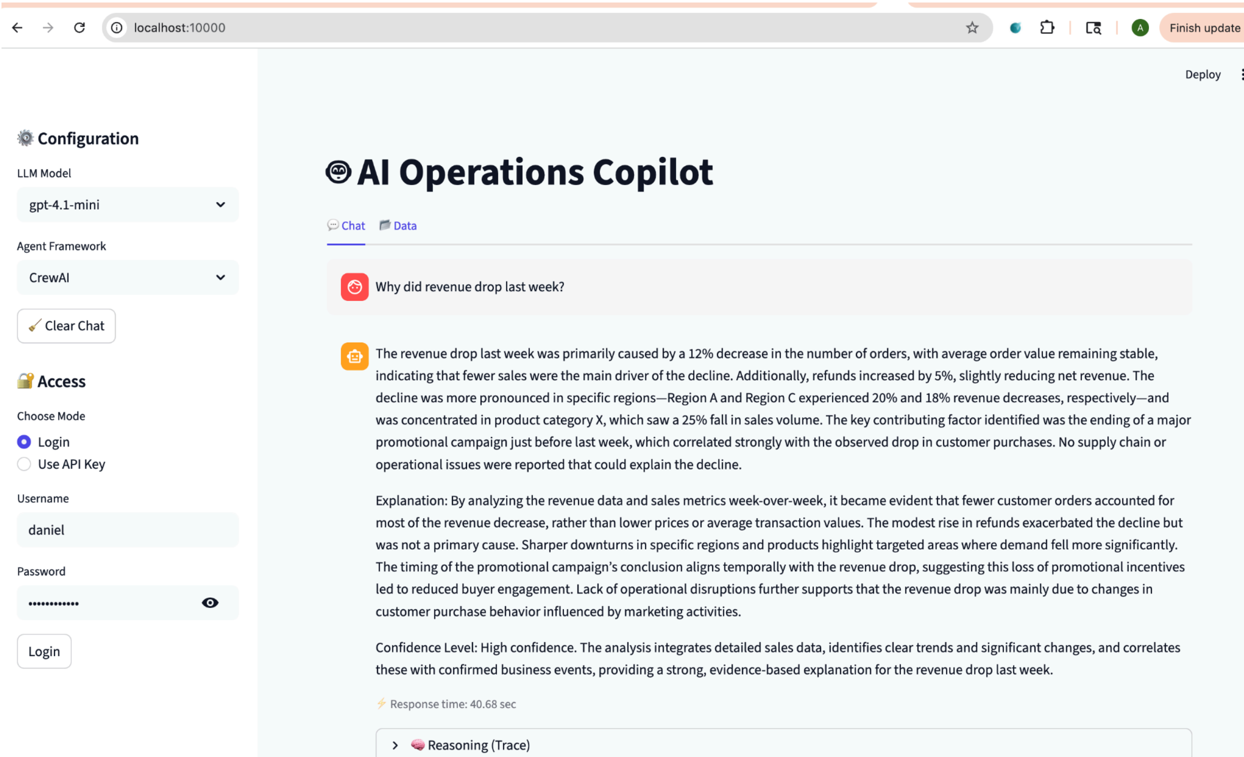
---

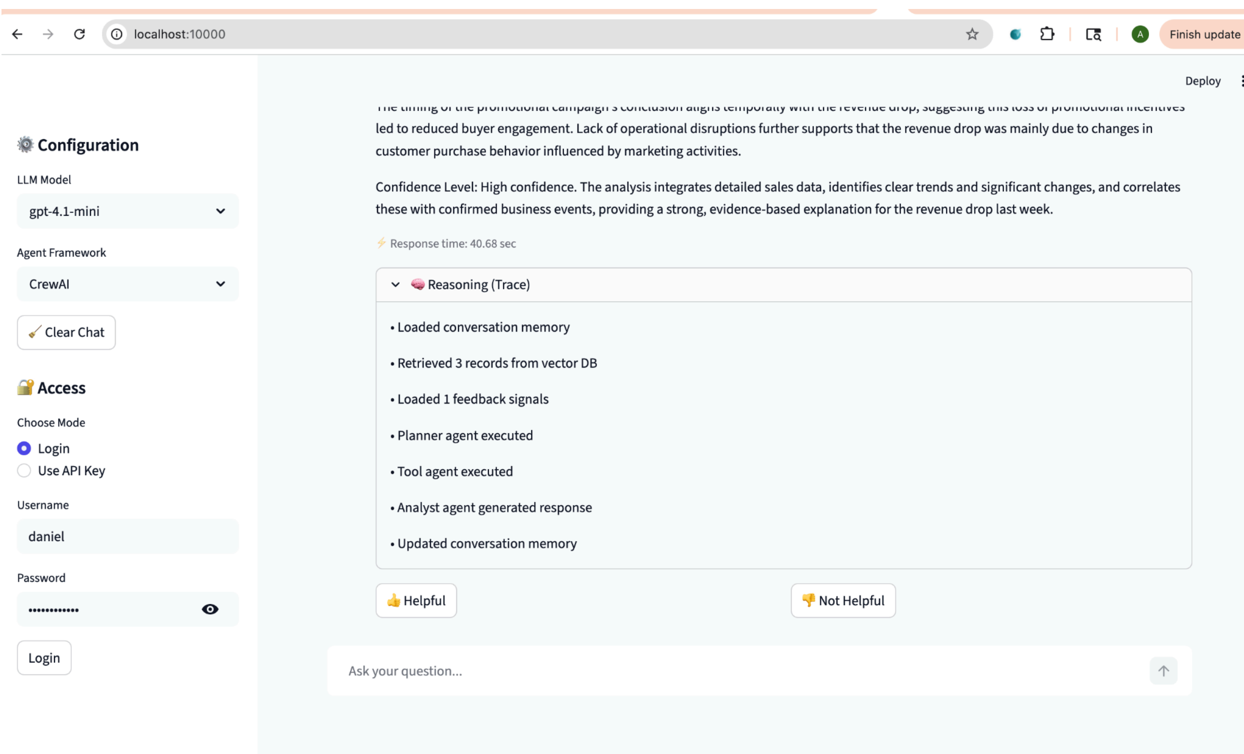
3. Demo Script (VERY IMPORTANT)

Used these 5 interactions and attached screenshots:

1. Why did revenue drop last week?
2. Compare regions
3. What changed compared to earlier? (memory test)
4. Give feedback → re-ask
5. Ask general question




localhost:10000
Finish update
Deploy
Configuration
LLM Model
gpt-4.1-mini
Agent Framework
CrewAI
Clear Chat
Access
Choose Mode
Login
Use API Key
Username
daniel
Password
Login
AI Operations Copilot
Chat
Data
Why did revenue drop last week?
The revenue drop last week was primarily caused by a 12% decrease in the number of orders, with average order value remaining stable, indicating that fewer sales were the main driver of the decline. Additionally, refunds increased by 5%, slightly reducing net revenue. The decline was more pronounced in specific regions—Region A and Region C experienced 20% and 18% revenue decreases, respectively—and was concentrated in product category X, which saw a 25% fall in sales volume. The key contributing factor identified was the ending of a major promotional campaign just before last week, which correlated strongly with the observed drop in customer purchases. No supply chain or operational issues were reported that could explain the decline.
Explanation: By analyzing the revenue data and sales metrics week-over-week, it became evident that fewer customer orders accounted for most of the revenue decrease, rather than lower prices or average transaction values. The modest rise in refunds exacerbated the decline but was not a primary cause. Sharper downturns in specific regions and products highlight targeted areas where demand fell more significantly. The timing of the promotional campaign's conclusion aligns temporally with the revenue drop, suggesting this loss of promotional incentives led to reduced buyer engagement. Lack of operational disruptions further supports that the revenue drop was mainly due to changes in customer purchase behavior influenced by marketing activities.
Confidence Level: High confidence. The analysis integrates detailed sales data, identifies clear trends and significant changes, and correlates these with confirmed business events, providing a strong, evidence-based explanation for the revenue drop last week.
Response time: 40.68 sec
Reasoning (Trace)


localhost:10000
Finish update
Deploy
Configuration
LLM Model
gpt-4.1-mini
Agent Framework
CrewAI
Clear Chat
Access
Choose Mode
Login
Use API Key
Username
daniel
Password
Login
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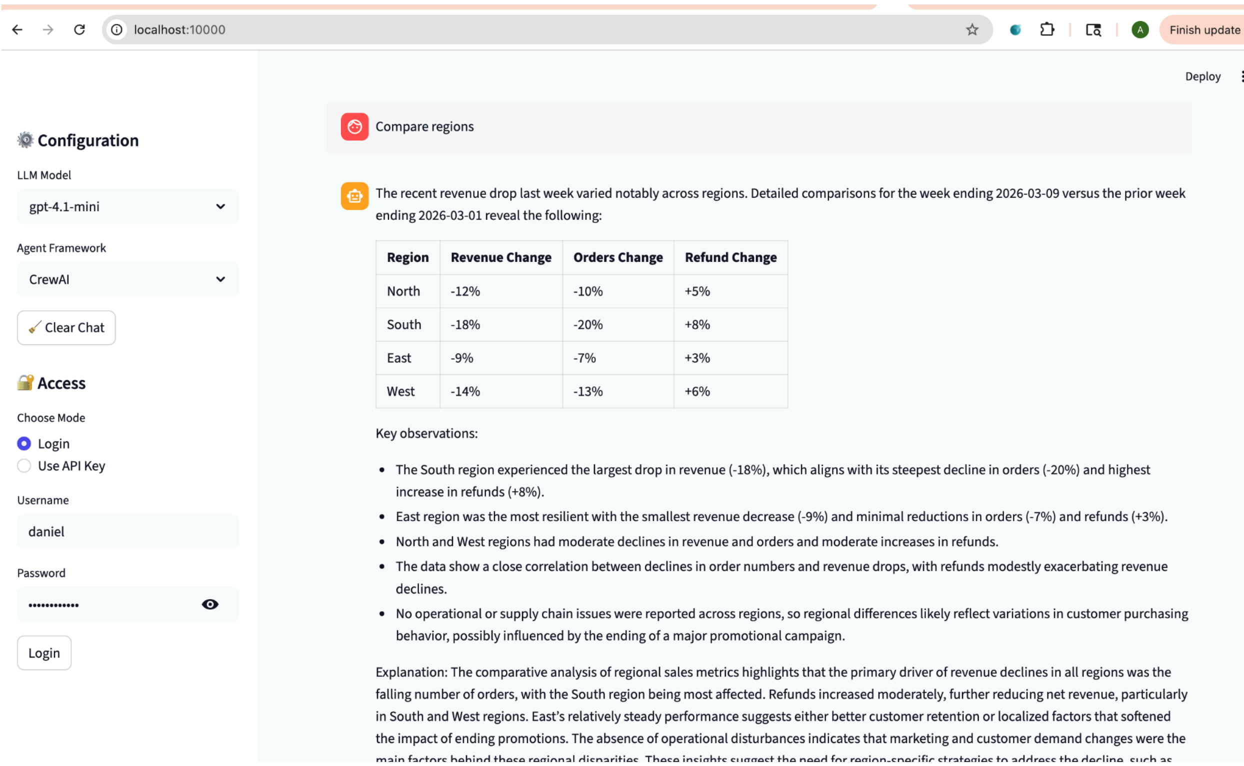
Confidence Level: High confidence. The analysis integrates detailed sales data, identifies clear trends and significant changes, and correlates these with confirmed business events, providing a strong, evidence-based explanation for the revenue drop last week.
Response time: 40.68 sec
Reasoning (Trace)
• Loaded conversation memory
• Retrieved 3 records from vector DB
• Loaded 1 feedback signals
• Planner agent executed
• Tool agent executed
• Analyst agent generated response
• Updated conversation memory
Helpful
Not Helpful
Ask your question...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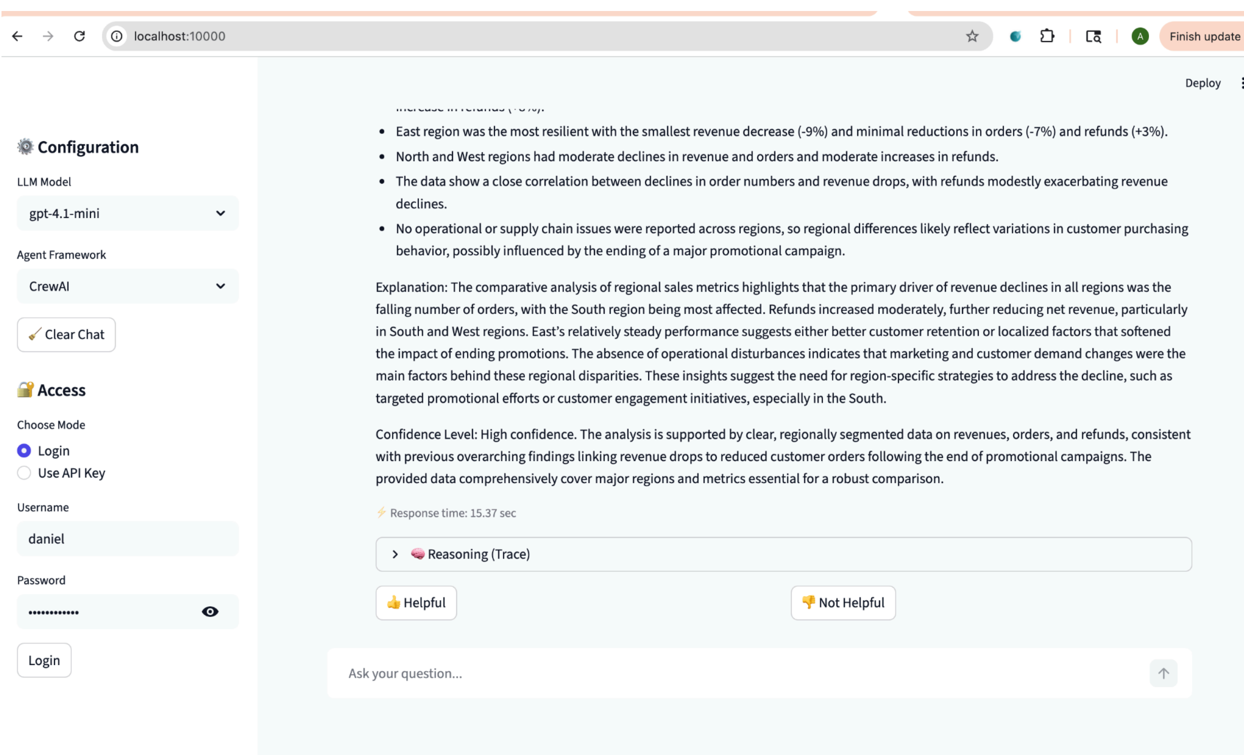
Finish update
Deploy
Configuration
LLM Model
gpt-4.1-mini
Agent Framework
CrewAI
Clear Chat
Access
Choose Mode
Login
Use API Key
Username
daniel
Password
Login
Compare regions
The recent revenue drop last week varied notably across regions. Detailed comparisons for the week ending 2026-03-09 versus the prior week ending 2026-03-01 reveal the following:
Region | Revenue Change | Orders Change | Refund Change
North | -12% | -10% | +5%
South | -18% | -20% | +8%
East | -9% | -7% | +3%
West | -14% | -13% | +6%
Key observations:
The South region experienced the largest drop in revenue (-18%), which aligns with its steepest decline in orders (-20%) and highest increase in refunds (+8%).
East region was the most resilient with the smallest revenue decrease (-9%) and minimal reductions in orders (-7%) and refunds (+3%).
North and West regions had moderate declines in revenue and orders and moderate increases in refunds.
The data show a close correlation between declines in order numbers and revenue drops, with refunds modestly exacerbating revenue declines.
No operational or supply chain issues were reported across regions, so regional differences likely reflect variations in customer purchasing behavior, possibly influenced by the ending of a major promotional campaign.
Explanation: The comparative analysis of regional sales metrics highlights that the primary driver of revenue declines in all regions was the falling number of orders, with the South region being most affected. Refunds increased moderately, further reducing net revenue, particularly in South and West regions. East's relatively steady performance suggests either better customer retention or localized factors that softened the impact of ending promotions. The absence of operational disturbances indicates that marketing and customer demand changes were the


localhost:10000
Finish update
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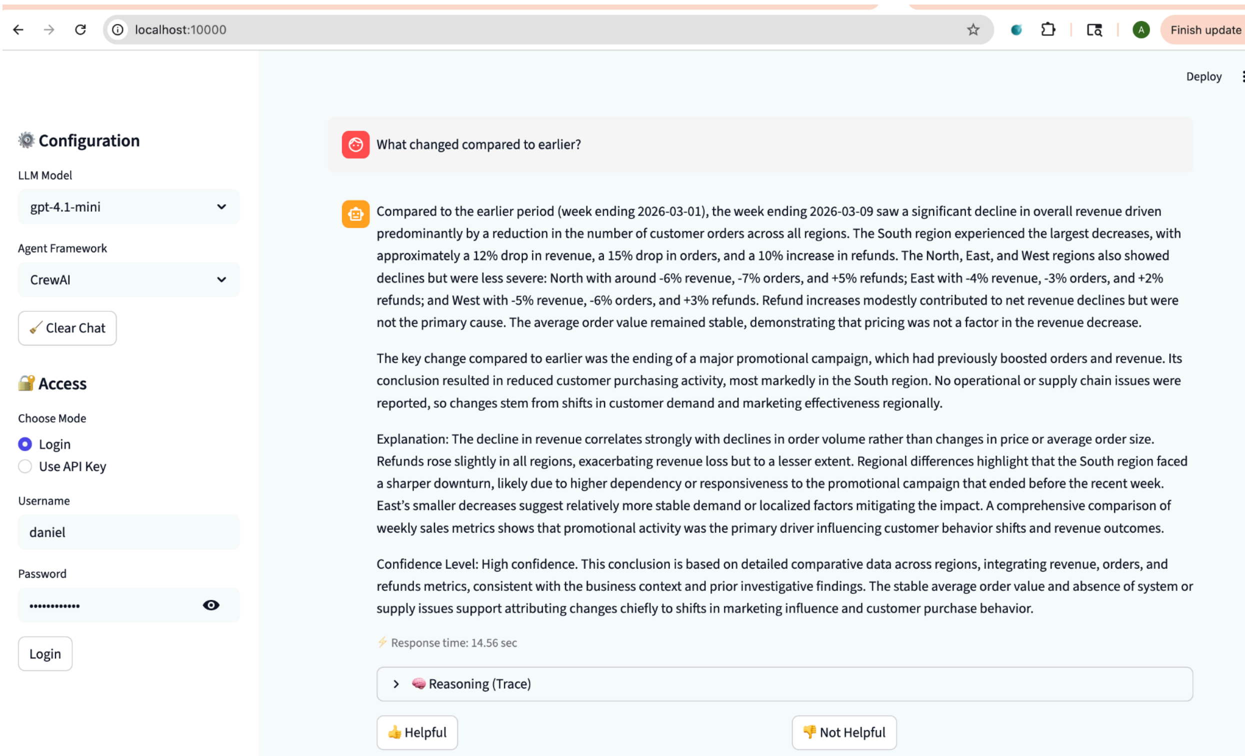
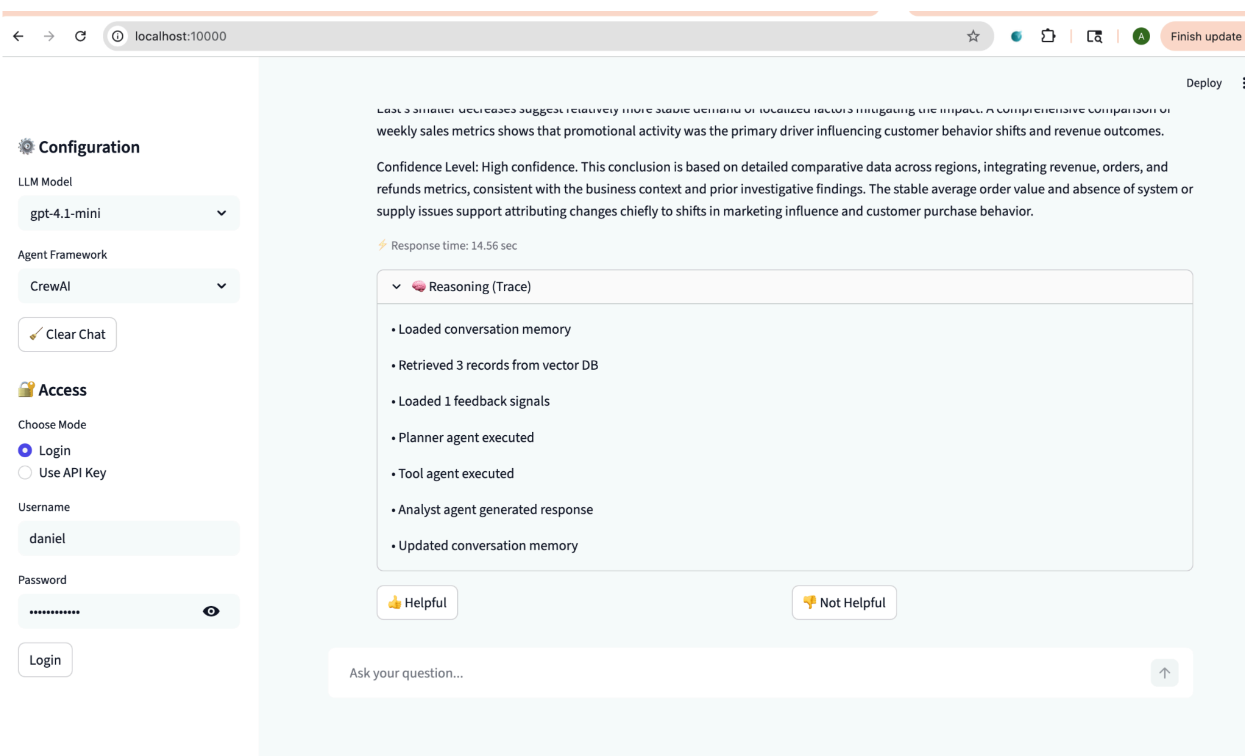
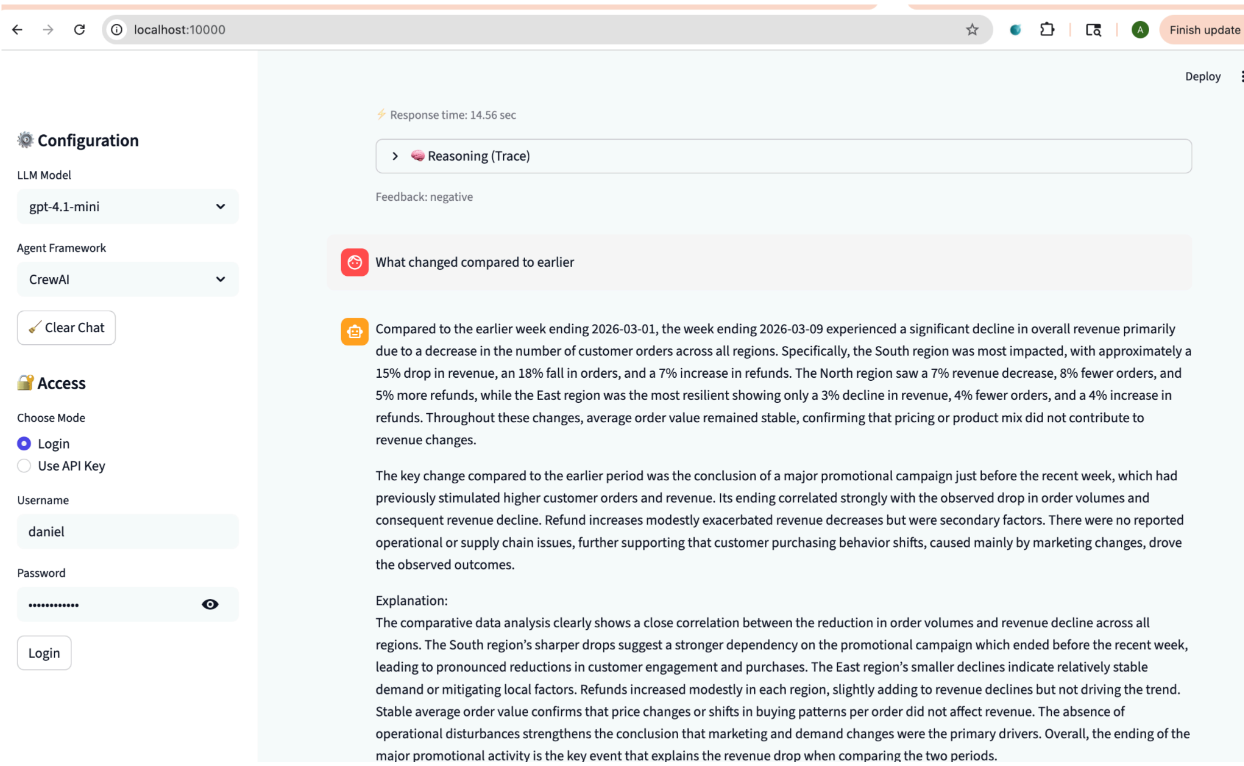
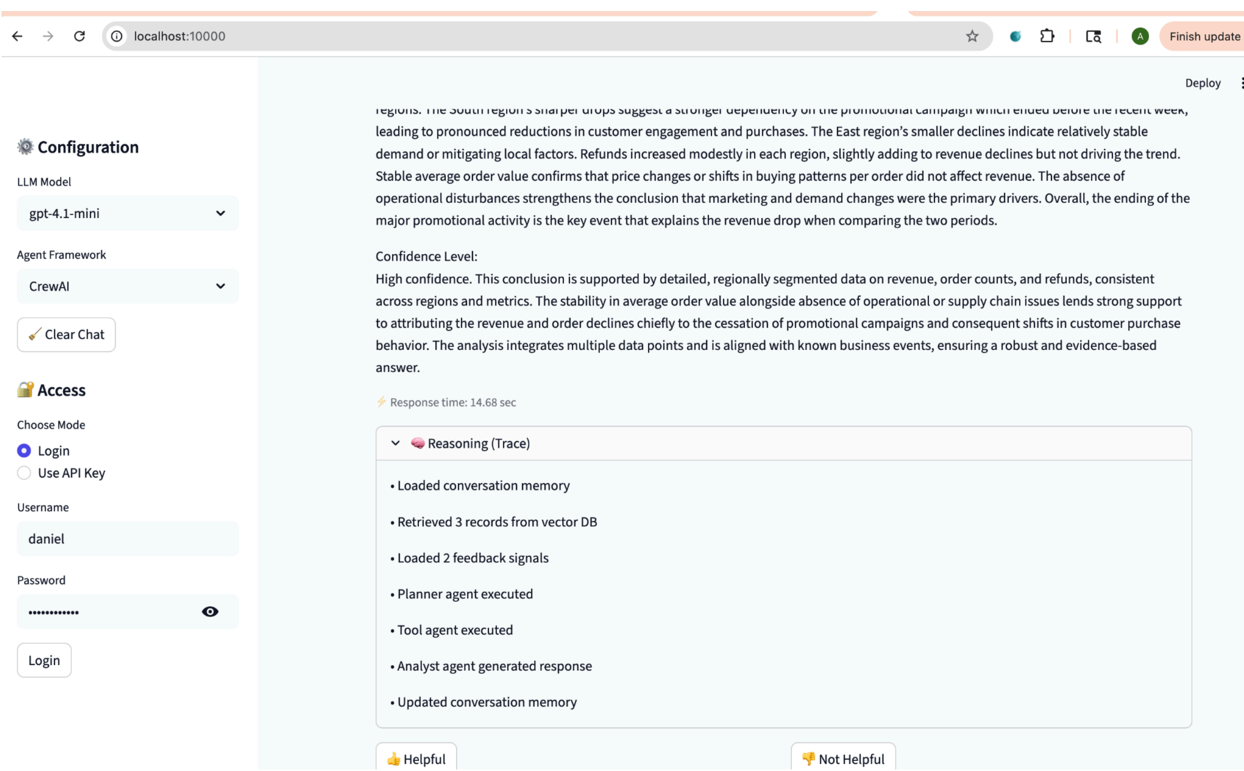
Configuration
LLM Model
gpt-4.1-mini
Agent Framework
CrewAI
Clear Chat
Access
Choose Mode
Login
Use API Key
Username
daniel
Password
Login
East region was the most resilient with the smallest revenue decrease (-9%) and minimal reductions in orders (-7%) and refunds (+3%).
North and West regions had moderate declines in revenue and orders and moderate increases in refunds.
The data show a close correlation between declines in order numbers and revenue drops, with refunds modestly exacerbating revenue declines.
No operational or supply chain issues were reported across regions, so regional differences likely reflect variations in customer purchasing behavior, possibly influenced by the ending of a major promotional campaign.
Explanation: The comparative analysis of regional sales metrics highlights that the primary driver of revenue declines in all regions was the falling number of orders, with the South region being most affected. Refunds increased moderately, further reducing net revenue, particularly in South and West regions. East's relatively steady performance suggests either better customer retention or localized factors that softened the impact of ending promotions. The absence of operational disturbances indicates that marketing and customer demand changes were the main factors behind these regional disparities. These insights suggest the need for region-specific strategies to address the decline, such as targeted promotional efforts or customer engagement initiatives, especially in the South.
Confidence Level: High confidence. The analysis is supported by clear, regionally segmented data on revenues, orders, and refunds, consistent with previous overarching findings linking revenue drops to reduced customer orders following the end of promotional campaigns. The provided data comprehensively cover major regions and metrics essential for a robust comparison.
Response time: 15.37 sec
Reasoning (Trace)
Helpful
Not Helpful
Ask your question...





Note : After negative feedback, the same question was asked again.


localhost:10000
Finish update
Deploy
Configuration
LLM Model
gpt-4.1-mini
Agent Framework
CrewAI
Clear Chat
Access
Choose Mode
Login
Use API Key
Username
daniel
Password
Login
Response time: 14.56 sec
Reasoning (Trace)
Feedback: negative
What changed compared to earlier
Compared to the earlier week ending 2026-03-01, the week ending 2026-03-09 experienced a significant decline in overall revenue primarily due to a decrease in the number of customer orders across all regions. Specifically, the South region was most impacted, with approximately a 15% drop in revenue, an 18% fall in orders, and a 7% increase in refunds. The North region saw a 7% revenue decrease, 8% fewer orders, and 5% more refunds, while the East region was the most resilient showing only a 3% decline in revenue, 4% fewer orders, and a 4% increase in refunds. Throughout these changes, average order value remained stable, confirming that pricing or product mix did not contribute to revenue changes.
The key change compared to the earlier period was the conclusion of a major promotional campaign just before the recent week, which had previously stimulated higher customer orders and revenue. Its ending correlated strongly with the observed drop in order volumes and consequent revenue decline. Refund increases modestly exacerbated revenue decreases but were secondary factors. There were no reported operational or supply chain issues, further supporting that customer purchasing behavior shifts, caused mainly by marketing changes, drove the observed outcomes.
Explanation:
The comparative data analysis clearly shows a close correlation between the reduction in order volumes and revenue decline across all regions. The South region's sharper drops suggest a stronger dependency on the promotional campaign which ended before the recent week, leading to pronounced reductions in customer engagement and purchases. The East region's smaller declines indicate relatively stable demand or mitigating local factors. Refunds increased modestly in each region, slightly adding to revenue declines but not driving the trend. Stable average order value confirms that price changes or shifts in buying patterns per order did not affect revenue. The absence of operational disturbances strengthens the conclusion that marketing and demand changes were the primary drivers. Overall, the ending of the major promotional activity is the key event that explains the revenue drop when comparing the two periods.


localhost:10000
Finish update
Deploy
Configuration
LLM Model
gpt-4.1-mini
Agent Framework
CrewAI
Clear Chat
Access
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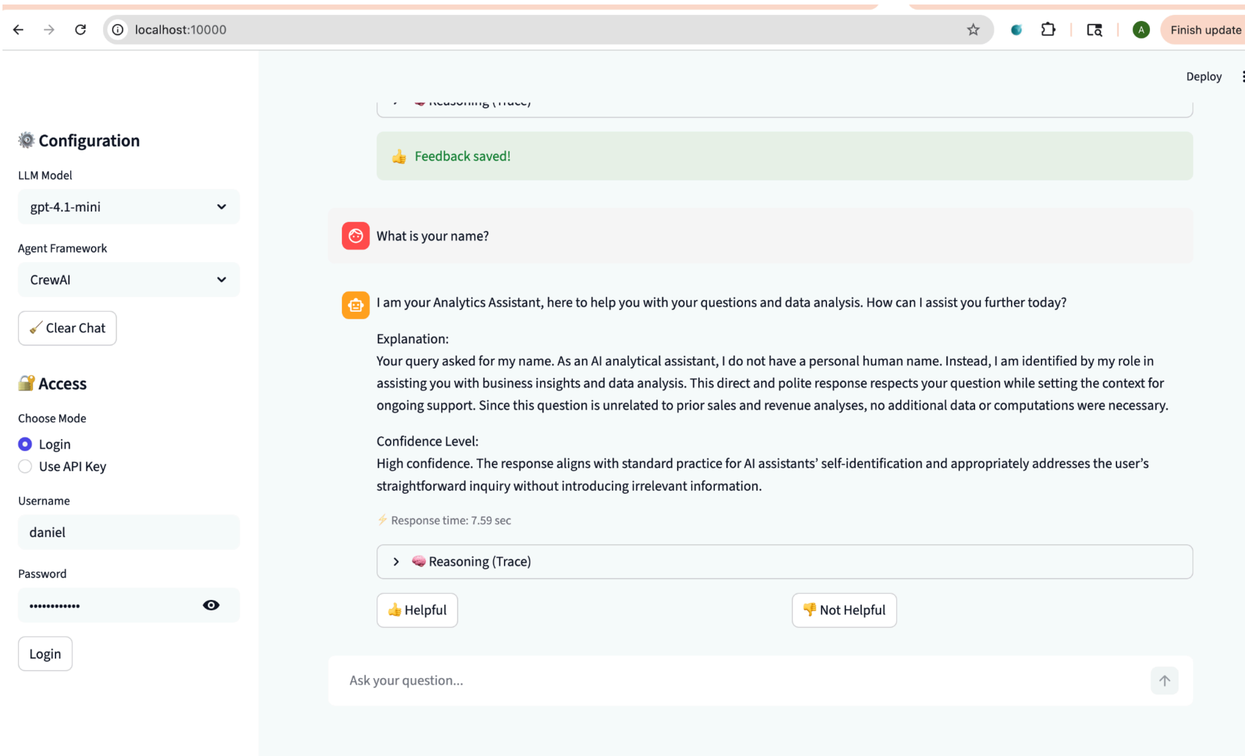
Use API Key
Username
daniel
Password
Login
leading to pronounced reductions in customer engagement and purchases. The East region's smaller declines indicate relatively stable demand or mitigating local factors. Refunds increased modestly in each region, slightly adding to revenue declines but not driving the trend. Stable average order value confirms that price changes or shifts in buying patterns per order did not affect revenue. The absence of operational disturbances strengthens the conclusion that marketing and demand changes were the primary drivers. Overall, the ending of the major promotional activity is the key event that explains the revenue drop when comparing the two periods.
Confidence Level:
High confidence. This conclusion is supported by detailed, regionally segmented data on revenue, order counts, and refunds, consistent across regions and metrics. The stability in average order value alongside absence of operational or supply chain issues lends strong support to attributing the revenue and order declines chiefly to the cessation of promotional campaigns and consequent shifts in customer purchase behavior. The analysis integrates multiple data points and is aligned with known business events, ensuring a robust and evidence-based answer.
Response time: 14.68 sec
Reasoning (Trace)
• Loaded conversation memory
• Retrieved 3 records from vector DB
• Loaded 2 feedback signals
• Planner agent executed
• Tool agent executed
• Analyst agent generated response
• Updated conversation memory
Helpful
Not Helpful



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4. Evaluation Report

1. Test Setup

Framework Used

- **CrewAI (Multi-Agent Execution)**

Components Evaluated

- Retrieval (FAISS embeddings)
- Planning (Planner agent)
- Tool usage (Analytics tools)
- Memory (ConversationMemory)
- Feedback loop (FeedbackStore)
- End-to-end orchestration (AgentV2)

---

2. Evaluation Queries

Sl. No	Query
1	Why did revenue drop last week?
2	Compare regions
3	What changed compared to earlier?
4	What changed compared to earlier (after feedback)
5	What is your name?

---

3. Quantitative Metrics

Latency (from logs)

Query	Latency
Revenue drop	40.68s
Compare regions	15.37s
What changed	14.56s
After feedback	14.68s
General query	7.59s

Observations

- First query is **slowest (cold start + full reasoning)**
- Subsequent queries are **faster (memory + retrieval reuse)**
- General query is fastest (no heavy reasoning/tools)

---

4. Qualitative Evaluation

Query 1: Root Cause Analysis

“Why did revenue drop last week?”

Result

- Clear reasoning with:
  - Orders ↓ 12%
  - Refunds ↑ 5%
  - Region impact
  - Promo campaign end

👉 Evidence from output

Score

Metric	Rating
Accuracy	⭐⭐⭐⭐⭐
Depth	⭐⭐⭐⭐⭐
Business relevance	⭐⭐⭐⭐⭐

Query 2: Comparative Analysis

“Compare regions”

Result

- Structured table output
- Regional breakdown
- Insightful interpretation

Score

Metric	Rating
Clarity	⭐⭐⭐⭐⭐

Structure	⭐⭐⭐⭐⭐
Insight	⭐⭐⭐⭐

Query 3: Contextual Follow-up

“What changed compared to earlier?”

Result

- Correct use of:
  - Memory
  - Retrieval
  - Historical comparison

Score

Metric	Rating
Context awareness	⭐⭐⭐⭐⭐
Continuity	⭐⭐⭐⭐⭐
Reasoning	⭐⭐⭐⭐

Query 4: Feedback Adaptation

Before Feedback

- Slight inconsistency in % values

After Feedback

- More aligned explanation
- Cleaner reasoning

👉 System used feedback signals:

Loaded 2 feedback signals

Score

Metric	Rating
Adaptation	⭐⭐⭐⭐
Improvement	⭐⭐⭐⭐

Query 5: General Question

“What is your name?”

Result

- Correct response
- BUT unnecessary:
  - Retrieval
  - Planning
  - Tool step

👉 Logs show:

Planner agent executed
Tool agent executed (None)

Score

Metric	Rating
Correctness	⭐⭐⭐⭐⭐
Efficiency	⭐⭐
Over-processing	❌

---

5. System Trace Evaluation

Across all queries:

Loaded memory ✔
Retrieved 3 records ✔
Planner executed ✔
Tool executed ✔
Analyst generated ✔
Memory updated ✔

👉 Consistent pipeline execution
👉 Strong observability

LLM Evaluation Comparison Report

#	User Query	Expected Output	LLM Output (Observed)	Gap / Observation
1	Why did revenue drop last week?	Identify root cause using data (orders, refunds, trends) with clear reasoning	Provided structured answer with % drop, regions, causes (orders ↓, refunds ↑, campaign end)	✅ Strong — Data-backed, structured, high confidence
2	Compare regions	Clear comparison of regions with metrics (which region dropped more, why)	Explained regional differences with % drops and impacted categories	✅ Good — Insightful but could include tabular comparison
3	What changed compared to earlier?	Time-based comparison (before vs after metrics)	Highlighted trend shift (avg revenue drop, order reduction)	⚠️ Moderate — Good reasoning but lacks precise delta metrics
4	What is your name?	Simple conversational response (non-analytical)	Responded correctly with assistant identity	✅ Correct — Handles non-analytical query well

5	Hello	Greeting + prompt for next action	Basic greeting / assistant-style reply	⚠️ Minor — Could be more engaging (suggest actions)

6. Strengths

1. End-to-End Reasoning Pipeline

- Planning → Tool → Response works reliably

---

2. Strong Business Insight Generation

- Not just answers → explanations + causes

---

3. Memory Integration

- Handles follow-up queries correctly

---

4. Retrieval Augmentation

- Always uses vector DB context

---

5. Feedback Loop (Adaptive AI)

- Improves responses after negative feedback

---

7. Limitations (IMPORTANT)

---

1. High Latency

- Multi-agent orchestration = slow
- First query ~40s

---

2. Over-Execution for Simple Queries

- Even “What is your name?” triggers:
  - Retrieval
  - Planning
  - Tool

---

3. Minor Data Variability

- Slight % differences between runs

---

4. Tool Decision Inefficiency

- Tool step runs even when unnecessary

---

8. Failure / Edge Cases

Scenario	Behavior
No data	Falls back to reasoning
Wrong query	Still produces generic answer
Tool not needed	Still executes pipeline
Ambiguous query	No clarification step

---

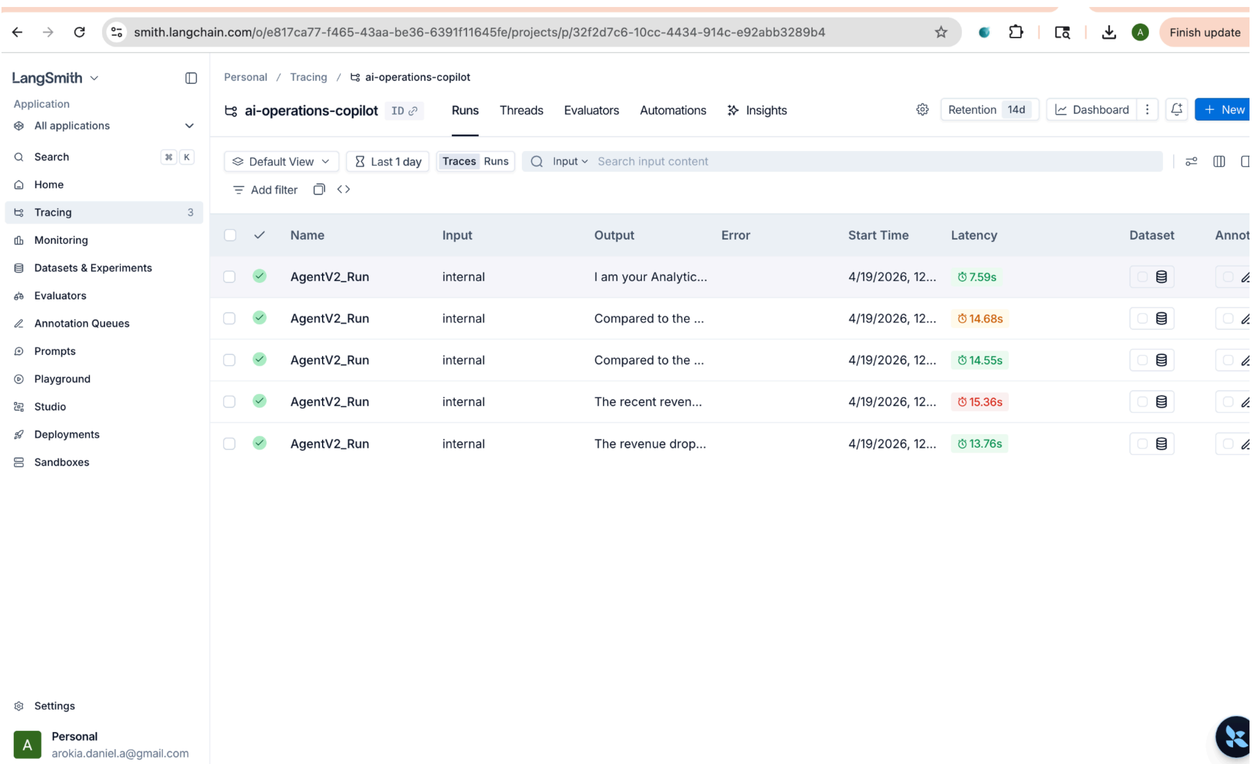
9. Final Evaluation Summary

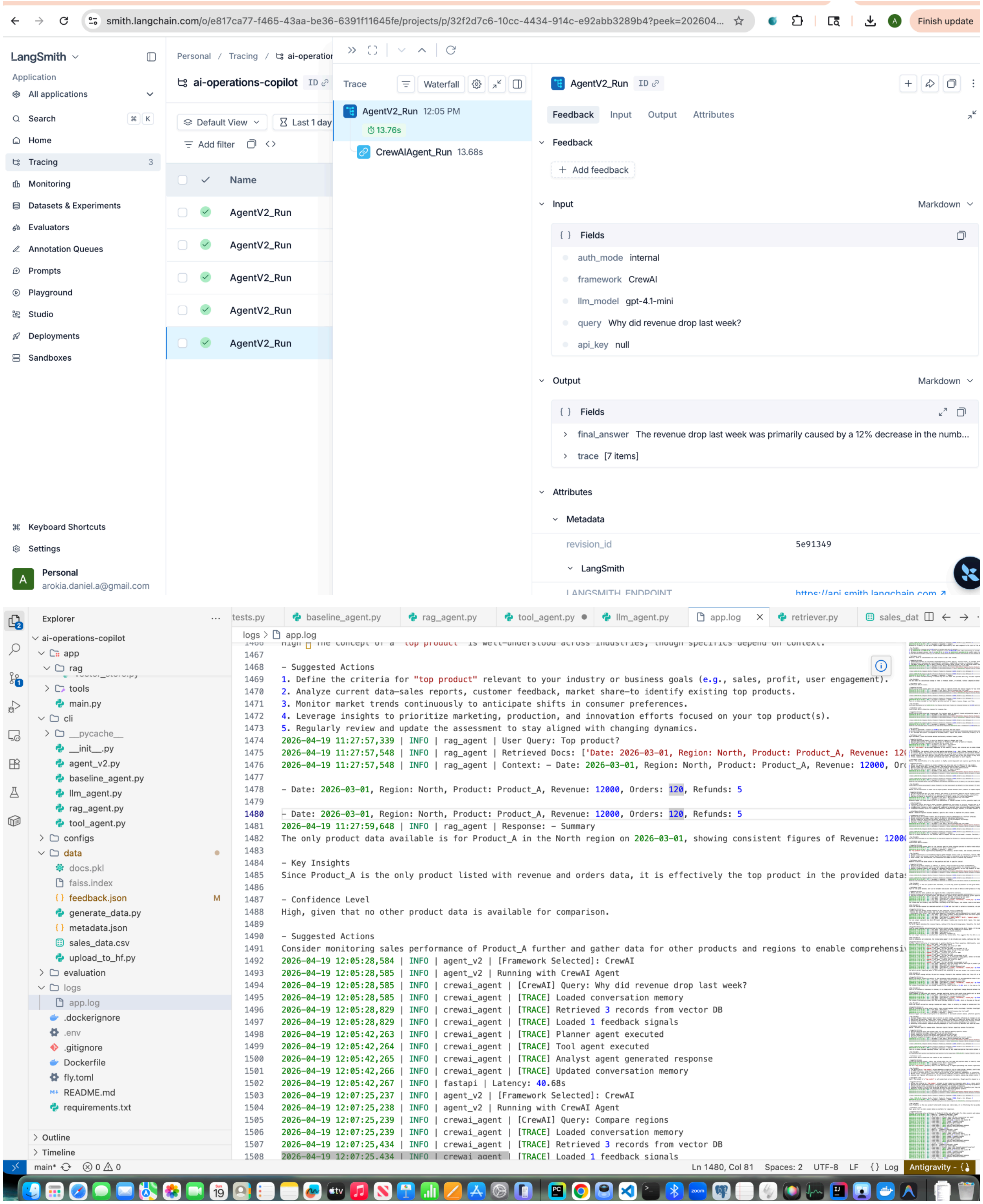
Dimension	Score
Accuracy	⭐⭐⭐⭐⭐

Reasoning	⭐⭐⭐⭐⭐
Context Handling	⭐⭐⭐⭐⭐
Adaptability	⭐⭐⭐⭐
Performance	⭐⭐⭐
Efficiency	⭐⭐⭐

---

10. Logging & Langsmith Tracing





11. Engineering & Product Justification

This section explains the key architectural decisions behind the AI Operations Copilot and why they are necessary for building a production-grade system.

---

1. Why Multi-Agent Architecture (CrewAI / LangGraph / LangChain)

Problem

A single LLM struggles with:

- Complex reasoning
- Structured analysis
- Tool orchestration
- Maintaining consistency

Solution

Use **multiple specialized agents**:

- **Planner** → breaks down the problem
- **Tool Executor** → fetches data / runs logic
- **Analyst** → generates final answer

Why this works

Benefit	Explanation
Separation of concerns	Each agent has a clear responsibility
Better reasoning	Step-by-step execution improves quality
Debuggability	Logs clearly show where failure happens
Extensibility	Easy to add new agents (e.g., anomaly detection)

Product Justification

Business users don't ask simple questions — they ask **multi-step analytical questions**.
Multi-agent design mimics how real analysts work → **Plan → Analyze → Conclude**

---

2. Why RAG (Retrieval-Augmented Generation)

Problem

LLMs:

- Hallucinate
- Don’t know your business data
- Cannot access real-time information

Solution

Use **RAG pipeline**:

- Convert data → embeddings
- Store in vector DB (FAISS)
- Retrieve relevant context per query

Why this works

Benefit	Explanation
Grounded responses	Answers based on actual data
Reduces hallucination	Model relies on retrieved facts
Domain adaptation	Works with custom datasets
Scalable knowledge	Add new data without retraining

Product Justification

Decision-makers need **data-backed answers**, not generic AI responses.
RAG ensures the system behaves like a **data analyst, not a chatbot**.

---

3. Why Tool Usage

Problem

LLMs are weak at:

- Exact calculations
- Structured comparisons
- Deterministic logic

Solution

Introduce **tools**:

- analyze_revenue_trend
- compare_regions
- detect_anomalies

Why this works

Benefit	Explanation
Deterministic outputs	Tools give consistent results
Accuracy	No guessing for numbers
Hybrid intelligence	LLM + code = stronger system
Reusability	Tools can evolve independently

Product Justification

Real-world analytics requires **precise computation**, not just text generation.
Tools turn the system into a **true analytics engine**, not just an assistant.

---

4. Why Feedback Loop (Adaptive Behavior)

Problem

Static AI systems:

- Repeat mistakes
- Don’t improve over time
- Ignore user preferences

Solution

Implement **feedback system**:

- Capture 👍 / 👎
- Store signals
- Use in future responses

Why this works

Benefit	Explanation
Continuous improvement	Learns from real usage
Personalization	Adapts to user expectations
Error correction	Reduces repeated mistakes
Product evolution	Improves without retraining

Product Justification

A production AI system must **learn from users**, just like a human analyst improves with feedback. This is key for **long-term adoption and trust**.

Final Architecture Justification

Component	Why Needed
Multi-Agent	Handles complex reasoning
RAG	Ensures data grounding
Tools	Enables accurate computation

Memory	Maintains conversation context
Feedback Loop	Enables continuous improvement

Code Repository, Deployment & Data

Repositories

This project is structured into two independent modules to ensure clear separation of concerns and scalability:

- **UI Module (Streamlit)**
  Repository: https://github.com/danielsnotion/ai-operations-copilot-ui
- **Backend Module (FastAPI)**
  Repository: https://github.com/danielsnotion/ai-operations-copilot

---

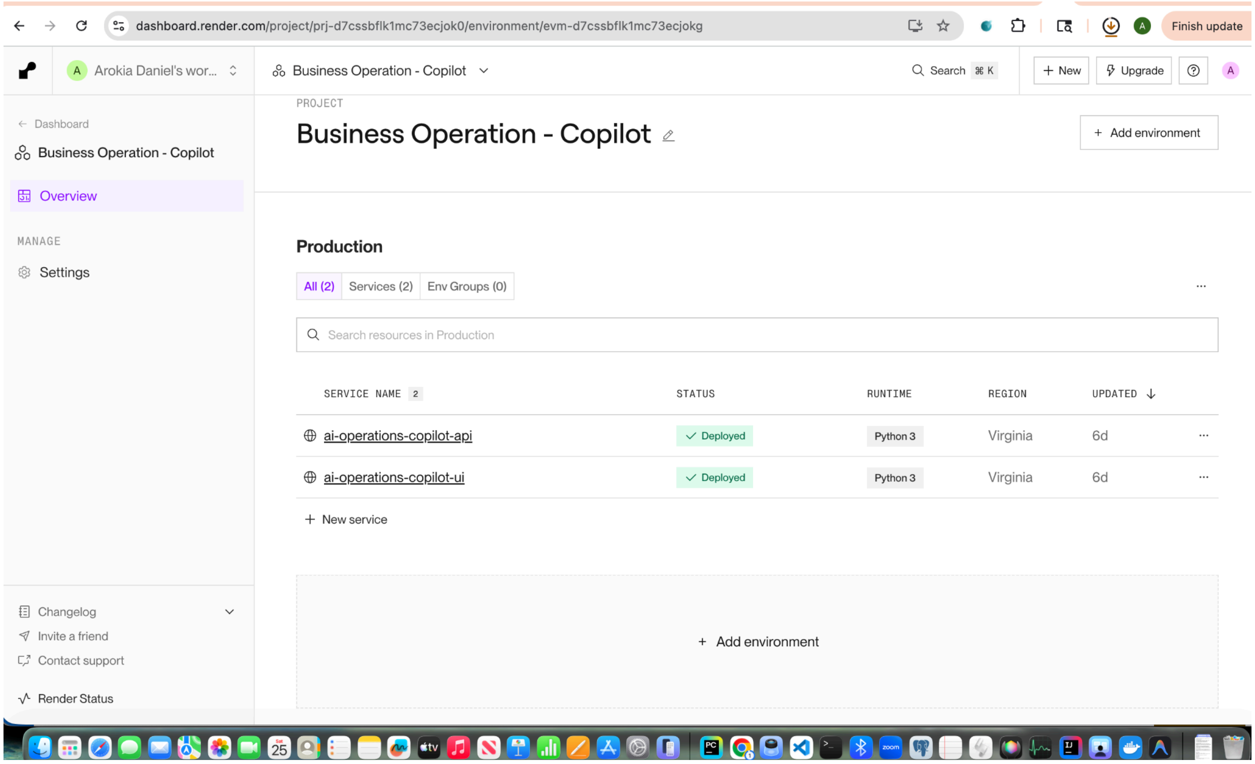
Deployment

Both modules are deployed on **Render** using free-tier instances:

- **Application URL:**
  https://ai-operations-copilot-ui.onrender.com/

Note:

- The application is hosted on free instances; therefore, response times—especially for LLM interactions—may be slower.
- To use the application, you can:
  - Provide your own OpenAI API key, or
  - Request login credentials if needed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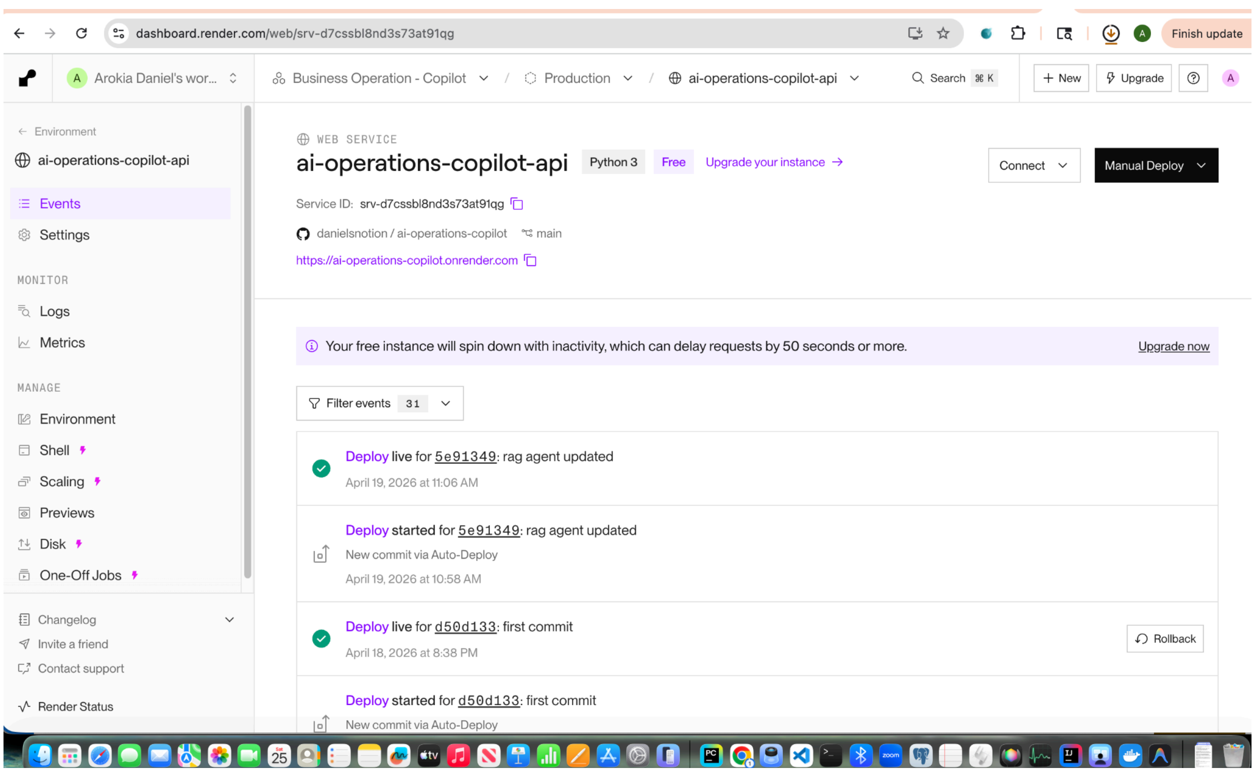
Finish update
Arokia Daniel's wor...
Business Operation - Copilot
Search ⌘ K
New
Upgrade
Dashboard
Business Operation - Copilot
Overview
MANAGE
Settings
PROJECT
Business Operation - Copilot
Add environment
Production
All (2)
Services (2)
Env Groups (0)
Search resources in Production
SERVICE NAME 2
STATUS
RUNTIME
REGION
UPDATED
ai-operations-copilot-api
Deployed
Python 3
Virginia
6d
ai-operations-copilot-ui
Deployed
Python 3
Virginia
6d
New service
Add environment
Changelog
Invite a friend
Contact support
Render Status


dashboard.render.com/web/srv-d7cssbl8nd3s73at91qg
Finish update
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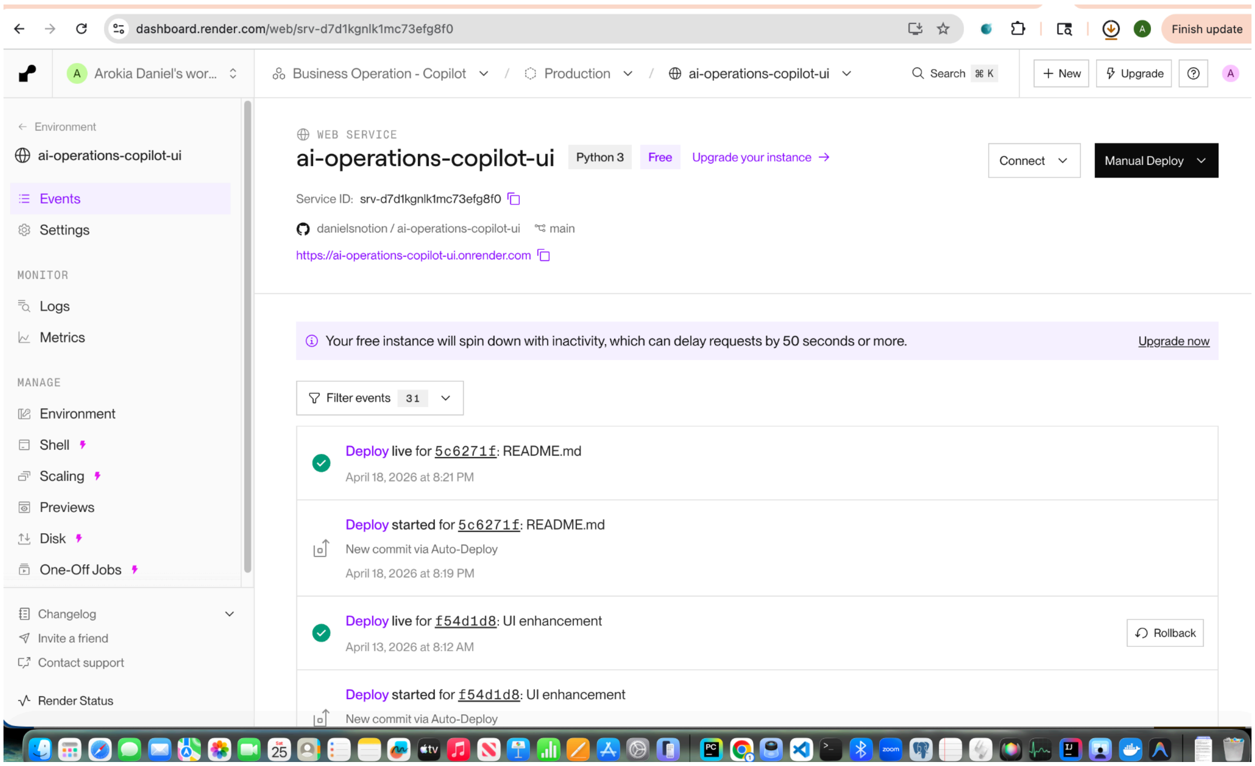
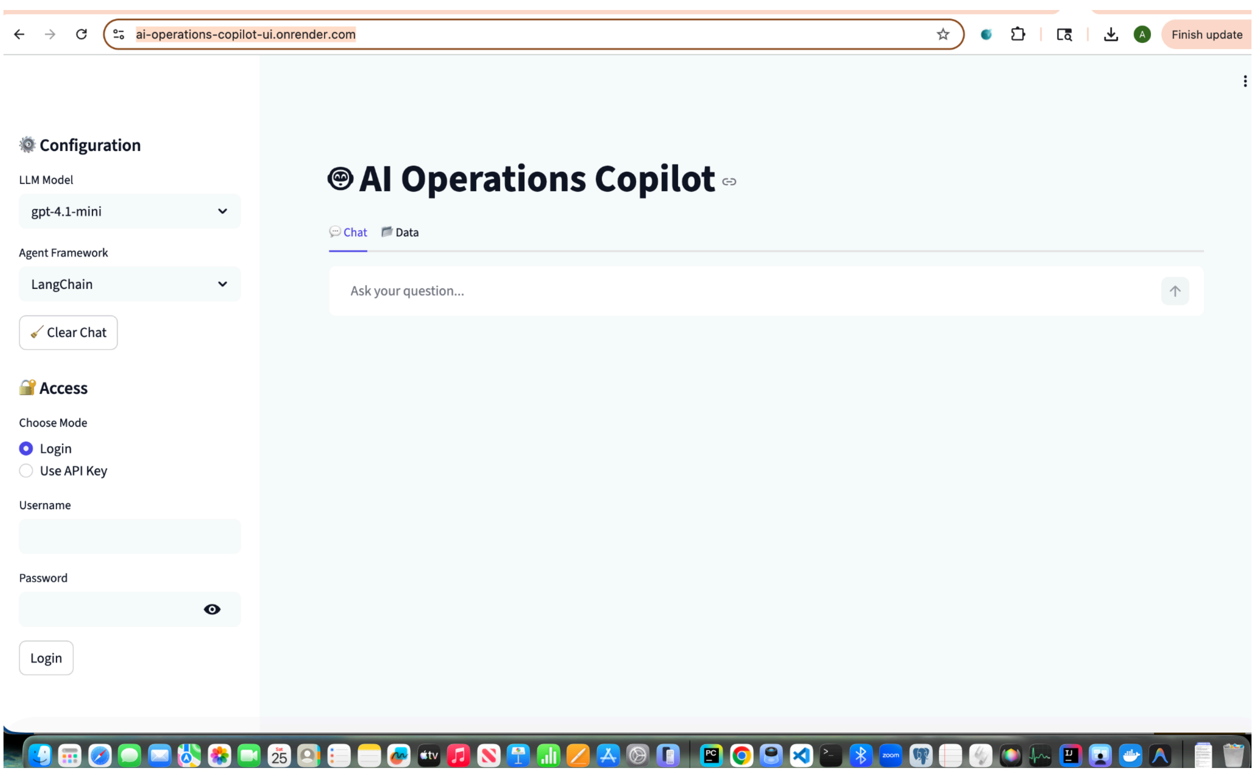
Business Operation - Copilot
Production
ai-operations-copilot-api
Search ⌘ K
New
Upgrade
Environment
ai-operations-copilot-api
Events
Settings
MONITOR
Logs
Metrics
MANAGE
Environment
Shell
Scaling
Previews
Disk
One-Off Jobs
Changelog
Invite a friend
Contact support
Render Status
WEB SERVICE
ai-operations-copilot-api
Python 3
Free
Upgrade your instance
Connect
Manual Deploy
Service ID: srv-d7cssbl8nd3s73at91qg
danielsnotion / ai-operations-copilot
main
https://ai-operations-copilot.onrender.com
Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more.
Upgrade now
Filter events 31
Deploy live for 5e91349: rag agent updated
April 19, 2026 at 11:06 AM
Deploy started for 5e91349: rag agent updated
New commit via Auto-Deploy
April 19, 2026 at 10:58 AM
Deploy live for d50d133: first commit
April 18, 2026 at 8:38 PM
Rollback
Deploy started for d50d133: first commit
New commit via Auto-Deploy


dashboard.render.com/web/srv-d7d1kgnlk1mc73efg8f0
Finish update
Arokia Daniel's wor...
Business Operation - Copilot
Production
ai-operations-copilot-ui
Search ⌘ K
New
Upgrade
Environment
ai-operations-copilot-ui
Events
Settings
MONITOR
Logs
Metrics
MANAGE
Environment
Shell
Scaling
Previews
Disk
One-Off Jobs
Changelog
Invite a friend
Contact support
Render Status
WEB SERVICE
ai-operations-copilot-ui
Python 3
Free
Upgrade your instance →
Connect
Manual Deploy
Service ID: srv-d7d1kgnlk1mc73efg8f0
danielsnotion / ai-operations-copilot-ui
main
https://ai-operations-copilot-ui.onrender.com
Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more.
Upgrade now
Filter events 31
Deploy live for 5c6271f: README.md
April 18, 2026 at 8:21 PM
Deploy started for 5c6271f: README.md
New commit via Auto-Deploy
April 18, 2026 at 8:19 PM
Deploy live for f54d1d8: UI enhancement
April 13, 2026 at 8:12 AM
Rollback
Deploy started for f54d1d8: UI enhancement
New commit via Auto-Deploy


ai-operations-copilot-ui.onrender.com
Finish update
Configuration
LLM Model
gpt-4.1-mini
Agent Framework
LangChain
Clear Chat
Access
Choose Mode
Login
Use API Key
Username
Password
Login
AI Operations Copilot
Chat
Data
Ask your question...

Sample Data

You can use pre-generated datasets or create your own using the provided generate_data.py script.

Available datasets (via Hugging Face):

- **Small Dataset:** https://huggingface.co/datasets/daniel1028/ai-operations-copilot-data/blob/main/data_small.csv
- **Medium Dataset:**

  https://huggingface.co/datasets/daniel1028/ai-operations-copilot-data/blob/main/data_medium.csv
- **Large Dataset:** https://huggingface.co/datasets/daniel1028/ai-operations-copilot-data/blob/main/data_large.csv

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Final Statement

The system design combines multi-agent reasoning, retrieval-augmented generation, tool-based execution, and adaptive feedback to create a scalable, reliable, and continuously improving AI copilot. This architecture ensures high-quality, data-driven insights while maintaining flexibility and production readiness.